# Online Appendix

When Does Crime Enter House Prices? Evidence from Chicago

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# A Data and Measurement

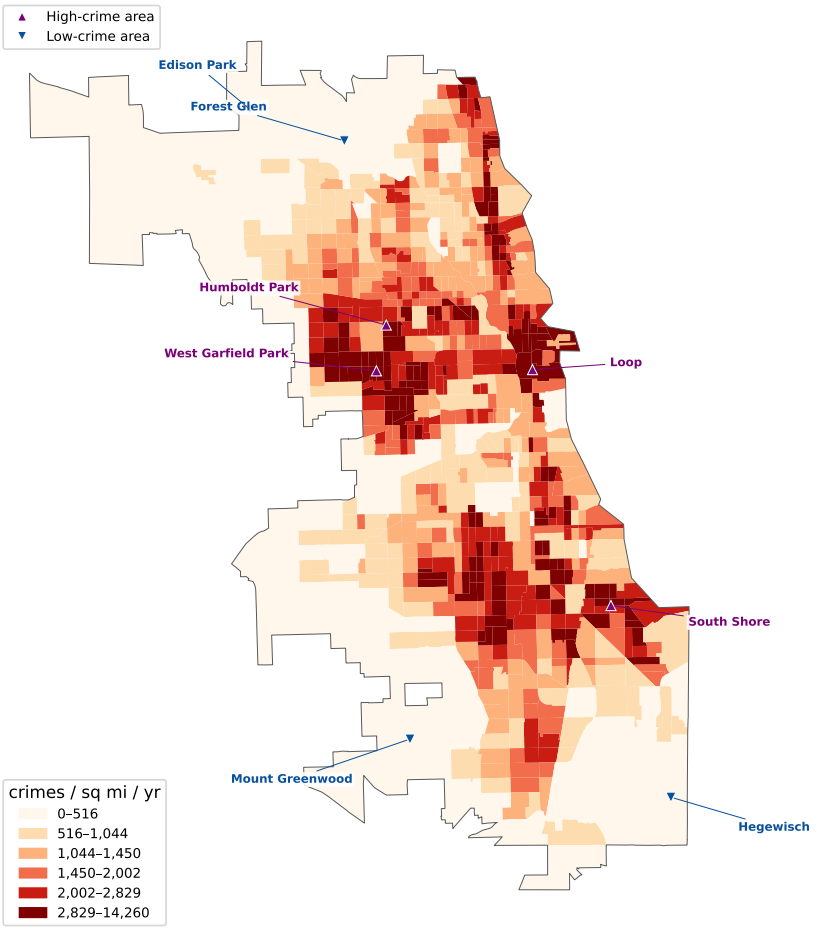



Figure OF1: Density of reported crime incidents, Chicago, 2007–2025.

*Notes:* Census-tract density of all reported Chicago Police Department incidents, 2007–2025, in incidents per square mile per year, binned into sextiles. Darker shades denote higher crime density.

Table OA1: Sample Construction

| | Transactions | Parcels |
|---|---|---|
| MLS sale listings (Chicago, SF/CO/RI) | 1,243,680 | |
| sold (closed transaction) | 562,770 | |
| closed 2007–2025 | 546,494 | |
| price \$10,000–\$10,000,000 | 543,535 | 341,896 |
| constant parcel count (analysis sample) | 530,395 | 337,461 |
| of which repeat-sale parcels ($\geq 2$) | 329,800 | 136,866 |
| identified by parcel + tract×month FE | 242,815 | |

*Notes:* Chicago Multiple Listing Service closed transactions in single-family, condominium, and residential-income properties. Within-property identification comes from the 136,866 parcels that transact at least twice. Adding tract×month fixed effects to the parcel fixed effects drops transactions that are the only sale in their tract×month cell (and the singletons this cascade creates), leaving the 242,815 transactions that identify the two-way fixed-effect model. Crime-specific specifications are smaller still (the near–far regressor also requires within-parcel variation in exposure), and every sale-price regression uses the 242,815-sale repeat-sale estimating sample unless a table states otherwise.

Table OA2: Listings With vs Without Nearby Crime

| | 3-month window | | 6-month window | |
|---|---|---|---|---|
| | No crime ($N = 97,348$) | Any crime ($N = 433,017$) | No crime ($N = 47,022$) | Any crime ($N = 483,343$) |
| Close price (\$) | 381,096 | 363,512 | 370,137 | 366,409 |
| Log close price | 12.54 | 12.42 | 12.53 | 12.43 |
| Days on market | 86 | 98 | 83 | 97 |
| Any list-price cut | 0.68 | 0.67 | 0.69 | 0.67 |
| Bedrooms | 3.1 | 3.0 | 3.0 | 3.0 |
| Bathrooms | 2.3 | 2.1 | 2.3 | 2.1 |
| Living area (sq. ft.) | 1,282 | 1,149 | 1,302 | 1,161 |
| Total rooms | 7.3 | 7.0 | 7.2 | 7.1 |
| Year built | 1954 | 1954 | 1955 | 1954 |

*Notes:* Sale analysis sample ($N = 530,395$; the 530,365 with an exposure classification are split) by whether any of the 27 offense groups records an incident within 0.05 miles in the 3 or 6 months before sale. Listings near recent crime are cheaper, smaller, and slower to sell; the raw price gap (Table 1) is confounded by these and unobserved differences, motivating the within-parcel, tract×month design.

Table OA3: White-Collar Offenses and the Financial-Magnitude Account

| Offense | Identifying parcels | Pooled 2007–25 | Crisis 2009–13 | Recovery 2014–25 |
|---|---|---|---|---|
| Fraud | 67,030 | −0.11 | −0.10 | −0.10 |
| | | (0.08) | (0.18) | (0.08) |
| Forgery | 23,614 | 0.21 | −0.04 | 0.16 |
| | | (0.17) | (0.66) | (0.17) |
| Bad checks | 8,919 | 0.66** | 1.60** | 0.42 |
| | | (0.33) | (0.75) | (0.47) |

*Notes.* White-collar offenses, estimated on the same specification as Table 2 ($\delta = (\beta_{\text{near}} - \beta_{\text{far}}) \times 100$; near 0–0.05 against 0.20–0.30 miles; parcel and tract × year-month fixed effects, clustered by tract; $N$ = 242,815). The financial-magnitude account predicts that offenses inflicting large monetary losses capitalize into prices. Embezzlement has too few incidents to estimate. Each offense is shown at its main window as in Table 2 (twelve months for offenses without a significant discount); identifying parcels are counted at that window; standard errors in parentheses.
$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

# B Additional Sale-Price Estimates

**Estimates as fixed effects are added.** Table OA4 reports the near–far estimate as the specification adds structure. Most of the raw association between near-ring crime and price reflects where and when properties sell: tract-by-month effects alone remove most of the raw homicide estimate. Observed characteristics are not a substitute for the parcel fixed effect. A cross-sectional hedonic regression overstates the discounts for burglary, narcotics, and vandalism, consistent with exposed properties differing in ways the listed characteristics do not record, and understates the homicide discount. Parcel fixed effects alone are not sufficient either. They absorb permanent differences across properties but not changes in the surrounding neighborhood between sales, and without tract-by-month effects the homicide estimate changes sign.

Table OA4: The Near–Far Discount Across Fixed-Effect Specifications

| Offense | (0) Raw | (1) Tract×mo. | (2) + Char. | (3) Parcel FE | (4) Parcel FE + tract×mo. |
|---|---|---|---|---|---|
| Homicide | $-12.81^{***}$ | $-2.29$ | $-2.63^{**}$ | 0.24 | $-3.02^{**}$ |
| | (3.71) | (1.43) | (1.07) | (1.63) | (1.38) |
| Burglary | $-3.50^{***}$ | $-0.71^{***}$ | $-0.32^{**}$ | $-0.90^{***}$ | $-0.24^{**}$ |
| | (0.50) | (0.27) | (0.15) | (0.21) | (0.12) |
| Narcotics | $-2.14^{***}$ | $-0.42^{***}$ | $-0.51^{***}$ | $-1.00^{***}$ | $-0.29^{**}$ |
| | (0.54) | (0.16) | (0.13) | (0.31) | (0.14) |
| Vandalism | $-3.52^{***}$ | $-1.18^{***}$ | $-0.48^{***}$ | $-0.31^{**}$ | $-0.17^{**}$ |
| | (0.59) | (0.22) | (0.13) | (0.15) | (0.08) |
| Tract×month FE | No | Yes | Yes | No | Yes |
| Characteristics | No | No | Yes | No | No |
| Parcel FE | No | No | No | Yes | Yes |

*Notes.* Each cell is a separate near–far regression of ln(sale price) on near (0–0.05 mi) and far (0.20–0.30 mi) incident counts; $\delta = (\beta_{\text{near}} - \beta_{\text{far}}) \times 100$ is the percent price change per near-ring incident, standard error below. Columns: (0) no fixed effects; (1) tract×year-month effects; (2) adds observed listing characteristics (bedrooms, bathrooms, ln living and lot area, age, units, stories, property type, new construction); (3) parcel fixed effects in place of characteristics; (4) parcel and tract×year-month fixed effects, the baseline specification. All five columns use the *same* rows, the column-(4) estimating sample ($N$ = 242,815 repeat-sale transactions); columns (1)–(4) also include the stale-crime persistence terms. Each offense is at its main window (homicide 10, burglary 8, narcotics 10, vandalism 6 months). Clustered by tract.
$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table OA5: Dose-Response for All Estimable Offenses

| Offense | 1 | 2 | 3 | 4+ | 1 | 2 | 3 | 4+ |
|---|---|---|---|---|---|---|---|---|
| | $\delta$: near incidents (vs. 0) | | | | listings identifying | | | |
| Homicide | $-2.22$ | $-20.17^{*}$ | 8.86 | 9.24 | 2,749 | 145 | 15 | 14 |
| | (1.43) | (10.72) | (27.74) | (6.06) | | | | |
| Burglary | $0.48^{*}$ | $-0.51$ | $-0.68$ | $-1.85^{**}$ | 55,036 | 20,475 | 7,903 | 6,598 |
| | (0.25) | (0.36) | (0.59) | (0.76) | | | | |
| Narcotics | $-0.34$ | 0.36 | $-2.06^{*}$ | $-0.74$ | 28,007 | 7,872 | 3,206 | 6,577 |
| | (0.36) | (0.69) | (1.12) | (1.24) | | | | |
| Vandalism | 0.21 | $-0.45$ | $-0.39$ | $-1.05^{**}$ | 62,935 | 28,364 | 13,049 | 12,067 |
| | (0.22) | (0.30) | (0.45) | (0.52) | | | | |

*Notes.* Price effect (%) of 1, 2, 3, and 4+ near (0–0.05 mi) incidents relative to zero, for all 22 estimable offenses. Parcel and tract × month fixed effects, pooled 2007–2025, clustered by tract; each offense at its main window (priced offenses) or a 12-month window otherwise. The final four columns ("listings") give the number of sale listings identifying each count (not parcels); "–" denotes no listings at that count.
$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table OA6: When the Discount Appears: Non-Cumulative Exposure

| Offense | Earliest sig. month | Late block (8–11 mo) | Concentrated in |
|---|---|---|---|
| Homicide | m1: $-5.86^{*}$ | $-5.95^{**}$ (2.91) | late (8–11 mo) |
| Burglary | m6: $-0.85^{***}$ | $-0.09$ (0.15) | early |
| Narcotics | m8: $-2.09^{***}$ | $-0.64^{**}$ (0.27) | late (8–11 mo) |
| Vandalism | m3: $-0.44^{**}$ | 0.12 (0.10) | early |

*Notes.* Non-cumulative decomposition of the near-ring discount by the lag at which crime occurs. "Earliest sig. month" is the first single month (crime in that month alone, $\delta = \beta_{\text{near},w} - \beta_{\text{near},w-1}$) at which the discount is significant; "Late block" is crime in the disjoint 8–11 month span entered as a single regressor (standard error in parentheses). Same specification as Table 2: parcel and tract × year-month fixed effects, clustered by tract, $N = 242{,}815$. Estimated jointly, the early and late blocks are statistically distinguishable only for vandalism (Figure A2); see Section 5.1.
$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table OA7: Sale-Price Capitalization by Domestic Status, All Offenses

| Window (months) | Dom $\delta_{\mathrm{D}}$ | Non-dom $\delta_{\mathrm{N}}$ | Wald $H_0\colon \delta_{\mathrm{D}} = \delta_{\mathrm{N}}$ |
|---|---|---|---|
| 1 | 0.20 | 0.05 | 0.634 |
| | (0.33) | (0.03) | |
| 2 | 0.05 | 0.03 | 0.933 |
| | (0.23) | (0.02) | |
| 3 | 0.00 | 0.03 | 0.885 |
| | (0.18) | (0.02) | |
| 4 | 0.03 | 0.01 | 0.904 |
| | (0.15) | (0.01) | |
| 5 | 0.06 | 0.00 | 0.646 |
| | (0.13) | (0.01) | |
| 6 | 0.01 | −0.01 | 0.910 |
| | (0.11) | (0.01) | |
| 7 | −0.02 | 0.00 | 0.882 |
| | (0.10) | (0.01) | |
| 8 | −0.06 | 0.00 | 0.547 |
| | (0.10) | (0.01) | |
| 9 | −0.14 | 0.00 | 0.143 |
| | (0.09) | (0.01) | |
| 10 | −0.13 | 0.00 | 0.145 |
| | (0.09) | (0.01) | |
| 11 | −0.17* | 0.00 | 0.062* |
| | (0.09) | (0.01) | |
| 12 | −0.17** | −0.01 | 0.046** |
| | (0.08) | (0.01) | |

*Notes.* Near–far price effect $\delta = (\beta_{\mathrm{near}} - \beta_{\mathrm{far}}) \times 100$, the percent sale-price change per near-ring incident (crime within 0–0.05 miles against 0.20–0.30 miles), at each cumulative exposure window from one to twelve months. Column $\delta_{\mathrm{D}}$ is the domestic-related arm (per the Illinois Domestic Violence Act) and $\delta_{\mathrm{N}}$ the non-domestic arm, both included in the same regression; the last column is the $p$-value of the Wald test that the two are equal. Incidents are pooled across the twenty-two named offense groups and split by the CPD domestic flag (1,003,504 domestic vs 4,357,006 non-domestic geocoded incidents, 2007–2025). Baseline specification: parcel and tract × year-month fixed effects, clustered by census tract; $N = 242{,}815$ repeat-sale transactions, pooled 2007–2025. Standard errors in parentheses. Incident counts come from the geocoded 2007–2025 records with the domestic flag and differ by about 1.4 percent from the incident counts in Table 1, which use the exposure-construction pool.
$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table OA8: Sale-Price Capitalization of Domestic versus Non-Domestic Assault

| Window (months) | Dom $\delta_{\mathrm{D}}$ | Non-dom $\delta_{\mathrm{N}}$ | Wald $H_0\colon \delta_{\mathrm{D}} = \delta_{\mathrm{N}}$ |
|---|---|---|---|
| 1 | 0.03 | −0.03 | 0.896 |
| | (0.37) | (0.20) | |
| 2 | 0.01 | −0.01 | 0.964 |
| | (0.26) | (0.14) | |
| 3 | −0.02 | 0.03 | 0.851 |
| | (0.21) | (0.11) | |
| 4 | 0.02 | −0.02 | 0.834 |
| | (0.17) | (0.09) | |
| 5 | 0.03 | −0.03 | 0.748 |
| | (0.15) | (0.08) | |
| 6 | 0.02 | −0.03 | 0.756 |
| | (0.13) | (0.07) | |
| 7 | −0.02 | −0.03 | 0.946 |
| | (0.12) | (0.07) | |
| 8 | −0.08 | −0.02 | 0.638 |
| | (0.12) | (0.06) | |
| 9 | −0.16 | −0.01 | 0.242 |
| | (0.11) | (0.05) | |
| 10 | −0.13 | 0.01 | 0.261 |
| | (0.10) | (0.05) | |
| 11 | −0.17 | −0.02 | 0.195 |
| | (0.10) | (0.04) | |
| 12 | −0.18* | −0.03 | 0.174 |
| | (0.10) | (0.04) | |

*Notes.* Near–far price effect $\delta = (\beta_{\mathrm{near}} - \beta_{\mathrm{far}}) \times 100$, the percent sale-price change per near-ring incident (crime within 0–0.05 miles against 0.20–0.30 miles), at each cumulative exposure window from one to twelve months. Domestic and non-domestic assault (split by the Chicago Police Department's domestic flag, based on the Illinois Domestic Violence Act) are separate near–far exposures in the same regression: column $\delta_{\mathrm{D}}$ is the domestic assault arm, $\delta_{\mathrm{N}}$ the non-domestic assault arm, and the last column the $p$-value of the Wald test that the two are equal. Parcel and tract × year-month fixed effects, clustered by census tract; $N$ = 242,815 repeat-sale transactions, pooled 2007–2025. Standard errors in parentheses.

$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table OA9: Listing-Agent Local Experience and the Asking Price, All Windows

| | **Exposure window (cumulative months)** | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Offense | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 |
| *Priced* | | | | | | | | | | | | |
| Homicide | −0.73 | −0.08 | −0.03 | −0.32 | −0.79* | −0.84** | −0.82** | −0.83** | −0.89** | −0.97*** | −0.88*** | −0.81*** |
| | (0.89) | (0.65) | (0.52) | (0.45) | (0.43) | (0.41) | (0.38) | (0.36) | (0.35) | (0.34) | (0.32) | (0.31) |
| Burglary | 0.03 | 0.02 | 0.02 | 0.05 | 0.06 | 0.01 | 0.03 | 0.00 | −0.01 | −0.01 | −0.02 | −0.00 |
| | (0.09) | (0.07) | (0.06) | (0.05) | (0.05) | (0.04) | (0.04) | (0.04) | (0.04) | (0.04) | (0.03) | (0.03) |
| Narcotics | −0.09 | −0.05 | −0.08 | −0.09* | −0.10* | −0.11** | −0.10*** | −0.07** | −0.07** | −0.08*** | −0.07*** | −0.07*** |
| | (0.12) | (0.08) | (0.07) | (0.05) | (0.05) | (0.05) | (0.04) | (0.03) | (0.03) | (0.03) | (0.03) | (0.02) |
| Vandalism | −0.02 | −0.04 | −0.03 | −0.03 | −0.04 | −0.03 | −0.02 | −0.03 | −0.04 | −0.05** | −0.06** | −0.06** |
| | (0.07) | (0.05) | (0.04) | (0.03) | (0.03) | (0.03) | (0.03) | (0.03) | (0.02) | (0.02) | (0.02) | (0.02) |
| *Suggestive* | | | | | | | | | | | | |
| Disorderly conduct | 0.12 | −0.08 | −0.04 | −0.22 | −0.13 | −0.12 | −0.07 | −0.05 | −0.05 | −0.08 | −0.07 | −0.07 |
| | (0.24) | (0.20) | (0.17) | (0.16) | (0.13) | (0.12) | (0.11) | (0.10) | (0.10) | (0.09) | (0.09) | (0.09) |
| Sex offenses | 0.12 | −0.01 | 0.03 | 0.07 | −0.02 | −0.00 | 0.01 | −0.03 | −0.05 | −0.06 | −0.08 | −0.08 |
| | (0.18) | (0.14) | (0.11) | (0.09) | (0.09) | (0.08) | (0.08) | (0.07) | (0.07) | (0.06) | (0.06) | (0.06) |
| *Precise nulls* | | | | | | | | | | | | |
| Theft | −0.01 | −0.01 | −0.01 | −0.01* | −0.00 | −0.00 | −0.00 | −0.00 | −0.00 | −0.00 | −0.00 | −0.01 |
| | (0.01) | (0.01) | (0.01) | (0.00) | (0.00) | (0.00) | (0.00) | (0.00) | (0.00) | (0.00) | (0.00) | (0.00) |
| Motor vehicle theft | 0.17** | 0.03 | 0.02 | 0.01 | 0.04 | 0.02 | 0.04 | 0.03 | 0.02 | 0.02 | 0.03 | 0.02 |
| | (0.08) | (0.06) | (0.05) | (0.04) | (0.03) | (0.03) | (0.03) | (0.03) | (0.03) | (0.03) | (0.02) | (0.02) |
| Robbery | −0.07 | −0.14 | −0.18** | −0.17** | −0.17*** | −0.19*** | −0.22*** | −0.20*** | −0.19*** | −0.19*** | −0.17*** | −0.16*** |
| | (0.11) | (0.09) | (0.08) | (0.07) | (0.06) | (0.06) | (0.05) | (0.05) | (0.05) | (0.04) | (0.04) | (0.04) |
| Assault (non-domestic) | −0.07 | −0.11*** | −0.10*** | −0.05* | −0.04 | −0.05** | −0.06*** | −0.07*** | −0.07*** | −0.08*** | −0.07*** | −0.07*** |
| | (0.05) | (0.04) | (0.03) | (0.03) | (0.02) | (0.02) | (0.02) | (0.02) | (0.02) | (0.02) | (0.02) | (0.01) |
| Assault (domestic) | −0.25*** | −0.15*** | −0.15*** | −0.11*** | −0.07* | −0.04 | −0.05 | −0.05 | −0.07** | −0.08*** | −0.06** | −0.06** |
| | (0.08) | (0.06) | (0.05) | (0.04) | (0.04) | (0.03) | (0.03) | (0.03) | (0.03) | (0.03) | (0.03) | (0.03) |
| Criminal trespass | 0.02 | −0.01 | −0.03 | 0.02 | 0.03 | 0.01 | −0.01 | −0.00 | 0.02 | 0.01 | 0.02 | 0.01 |
| | (0.11) | (0.07) | (0.05) | (0.05) | (0.04) | (0.04) | (0.04) | (0.04) | (0.03) | (0.03) | (0.03) | (0.03) |
| Fraud | −0.07 | −0.08* | −0.05 | −0.09** | −0.08** | −0.07** | −0.07*** | −0.07*** | −0.05** | −0.05** | −0.04** | −0.04** |
| | (0.07) | (0.05) | (0.04) | (0.04) | (0.03) | (0.03) | (0.03) | (0.03) | (0.02) | (0.02) | (0.02) | (0.02) |
| Prostitution | −0.21 | −0.03 | 0.08 | 0.08 | 0.10 | 0.07 | −0.02 | −0.03 | −0.03 | −0.02 | 0.00 | −0.03 |
| | (0.24) | (0.18) | (0.14) | (0.10) | (0.08) | (0.07) | (0.06) | (0.05) | (0.04) | (0.03) | (0.04) | (0.03) |
| Forgery | −0.06 | −0.19 | −0.20* | −0.16 | −0.13 | −0.14 | −0.14 | −0.12 | −0.13 | −0.15* | −0.16* | −0.14 |
| | (0.19) | (0.13) | (0.12) | (0.11) | (0.10) | (0.10) | (0.10) | (0.10) | (0.09) | (0.09) | (0.08) | (0.09) |
| Family offenses | −0.15 | 0.00 | −0.15 | −0.20 | −0.19 | −0.18 | −0.24 | −0.19 | −0.15 | −0.17 | −0.09 | −0.07 |
| | (0.44) | (0.29) | (0.24) | (0.22) | (0.19) | (0.18) | (0.17) | (0.16) | (0.16) | (0.16) | (0.15) | (0.15) |
| Bad checks | −0.50 | −0.16 | −0.08 | −0.15 | −0.14 | −0.23 | −0.29** | −0.26** | −0.22* | −0.17* | −0.14 | −0.15 |
| | (0.31) | (0.23) | (0.20) | (0.18) | (0.17) | (0.15) | (0.13) | (0.13) | (0.12) | (0.10) | (0.10) | (0.10) |
| Liquor law | −0.65* | −0.45* | −0.18 | −0.23 | −0.20 | −0.06 | −0.08 | −0.07 | −0.10 | −0.12 | −0.15 | −0.13 |
| | (0.37) | (0.26) | (0.21) | (0.19) | (0.17) | (0.15) | (0.15) | (0.14) | (0.13) | (0.13) | (0.12) | (0.11) |
| *Inconclusive* | | | | | | | | | | | | |
| Weapons | 0.14 | 0.13 | 0.21** | 0.15 | 0.17* | 0.12 | 0.11 | 0.11 | 0.09 | 0.13** | 0.12** | 0.11* |
| | (0.21) | (0.14) | (0.10) | (0.10) | (0.09) | (0.09) | (0.08) | (0.07) | (0.07) | (0.06) | (0.06) | (0.06) |
| Arson | 0.65 | 0.12 | 0.12 | 0.13 | −0.07 | −0.03 | 0.13 | −0.04 | 0.01 | 0.09 | 0.10 | 0.05 |
| | (0.74) | (0.60) | (0.52) | (0.42) | (0.40) | (0.38) | (0.34) | (0.33) | (0.32) | (0.31) | (0.30) | (0.29) |
| Gambling | −1.61 | −1.46 | −1.14 | −0.26 | −0.41 | −0.71 | −0.85 | −0.90 | −1.05* | −1.09** | −1.05** | −1.00** |
| | (1.58) | (1.02) | (0.82) | (0.80) | (0.69) | (0.69) | (0.65) | (0.63) | (0.57) | (0.54) | (0.50) | (0.49) |
| Stolen property | −0.26 | −0.18 | −0.68 | −0.95 | −0.82 | −0.65 | −0.66 | −0.67 | −0.42 | −0.51 | −0.51 | −0.33 |
| | (1.08) | (0.89) | (0.87) | (0.66) | (0.63) | (0.54) | (0.50) | (0.48) | (0.46) | (0.42) | (0.41) | (0.39) |

*Notes.* Local-experience gradient, (near × $E$ − far × $E$) × 100, at every cumulative window, where $E = \ln(1 +$ prior same-agent same-tract listings) is the listing agent's local experience, under the specification of Table 5: parcel, tract × month, and agent fixed effects; standard errors clustered by tract in parentheses; $N$ = 972,629. Offense groups as in Table A4. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table OA10: Buyer's-Agent Local Experience and the Closing Price, All Windows

| | **Exposure window (cumulative months)** | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Offense | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 |
| *Priced* | | | | | | | | | | | | |
| Homicide | −6.27** | −2.46 | −2.04* | −1.29 | −0.95 | −1.06 | −1.44* | −1.58** | −1.47** | −1.61** | −1.84** | −2.19*** |
| | (2.60) | (1.73) | (1.18) | (1.21) | (0.88) | (0.81) | (0.78) | (0.78) | (0.70) | (0.75) | (0.79) | (0.81) |
| Burglary | 0.53** | 0.22 | 0.05 | 0.02 | 0.12 | 0.10 | 0.08 | 0.09 | 0.10 | 0.15 | 0.13 | 0.12 |
| | (0.27) | (0.19) | (0.16) | (0.15) | (0.13) | (0.13) | (0.11) | (0.10) | (0.11) | (0.09) | (0.09) | (0.08) |
| Narcotics | −0.06 | 0.37 | 0.38* | 0.32* | 0.34** | 0.18 | 0.11 | 0.10 | 0.09 | 0.08 | 0.06 | 0.10 |
| | (0.47) | (0.29) | (0.22) | (0.19) | (0.16) | (0.14) | (0.13) | (0.11) | (0.09) | (0.10) | (0.10) | (0.08) |
| Vandalism | −0.04 | 0.09 | 0.06 | 0.06 | 0.11 | 0.14** | 0.13** | 0.09* | 0.10* | 0.09* | 0.04 | 0.03 |
| | (0.14) | (0.12) | (0.09) | (0.08) | (0.07) | (0.06) | (0.06) | (0.05) | (0.05) | (0.05) | (0.05) | (0.05) |
| *Suggestive* | | | | | | | | | | | | |
| Disorderly conduct | −0.15 | −0.11 | 0.06 | −0.18 | −0.33 | −0.42 | −0.44** | −0.24 | −0.34* | −0.48** | −0.42** | −0.47** |
| | (0.57) | (0.47) | (0.35) | (0.29) | (0.30) | (0.26) | (0.22) | (0.21) | (0.19) | (0.19) | (0.21) | (0.21) |
| Sex offenses | −0.49 | −0.20 | 0.08 | 0.18 | 0.30 | 0.19 | 0.15 | 0.16 | 0.15 | 0.09 | 0.03 | 0.03 |
| | (0.52) | (0.36) | (0.28) | (0.25) | (0.26) | (0.23) | (0.21) | (0.18) | (0.16) | (0.15) | (0.15) | (0.14) |
| *Precise nulls* | | | | | | | | | | | | |
| Theft | 0.02 | −0.01 | −0.01 | −0.01 | −0.00 | −0.00 | 0.01 | 0.01 | 0.01 | 0.00 | 0.01 | 0.00 |
| | (0.01) | (0.01) | (0.01) | (0.01) | (0.01) | (0.01) | (0.01) | (0.01) | (0.01) | (0.01) | (0.01) | (0.01) |
| Motor vehicle theft | 0.39* | 0.48*** | 0.32** | 0.32*** | 0.24** | 0.10 | 0.05 | 0.07 | 0.03 | 0.07 | 0.05 | 0.00 |
| | (0.22) | (0.14) | (0.13) | (0.12) | (0.10) | (0.09) | (0.09) | (0.07) | (0.07) | (0.07) | (0.07) | (0.06) |
| Robbery | 0.43 | 0.29 | 0.20 | 0.16 | 0.14 | 0.08 | −0.00 | 0.05 | −0.04 | 0.02 | 0.03 | 0.03 |
| | (0.28) | (0.20) | (0.17) | (0.16) | (0.11) | (0.12) | (0.10) | (0.10) | (0.09) | (0.08) | (0.07) | (0.07) |
| Assault (non-domestic) | 0.45*** | 0.14 | 0.14* | 0.12* | 0.13* | 0.16** | 0.10* | 0.11** | 0.06 | 0.04 | 0.04 | 0.01 |
| | (0.15) | (0.09) | (0.08) | (0.07) | (0.07) | (0.06) | (0.05) | (0.05) | (0.04) | (0.04) | (0.03) | (0.03) |
| Assault (domestic) | 0.15 | −0.12 | −0.18 | −0.14 | −0.13 | −0.13 | −0.05 | 0.01 | −0.01 | 0.02 | −0.01 | −0.05 |
| | (0.27) | (0.17) | (0.13) | (0.11) | (0.08) | (0.08) | (0.07) | (0.07) | (0.06) | (0.07) | (0.05) | (0.05) |
| Criminal trespass | 0.17 | 0.25 | 0.18 | 0.24** | 0.16 | 0.03 | 0.05 | 0.02 | −0.00 | −0.00 | 0.03 | 0.05 |
| | (0.25) | (0.16) | (0.12) | (0.10) | (0.11) | (0.11) | (0.10) | (0.11) | (0.10) | (0.09) | (0.07) | (0.06) |
| Fraud | −0.09 | −0.08 | −0.07 | −0.12 | −0.08 | −0.11 | −0.08 | 0.00 | 0.01 | 0.00 | 0.05 | 0.01 |
| | (0.32) | (0.13) | (0.09) | (0.08) | (0.09) | (0.10) | (0.08) | (0.07) | (0.05) | (0.05) | (0.04) | (0.04) |
| Prostitution | 1.37*** | 0.26 | −0.19 | −0.24 | −0.01 | −0.09 | −0.08 | −0.13 | −0.09 | −0.07 | −0.10 | −0.13 |
| | (0.43) | (0.19) | (0.19) | (0.15) | (0.14) | (0.22) | (0.17) | (0.15) | (0.11) | (0.10) | (0.10) | (0.08) |
| Forgery | −0.38 | −0.15 | 0.14 | 0.17 | 0.14 | 0.05 | −0.08 | −0.10 | −0.18 | −0.19 | −0.23* | −0.21* |
| | (0.58) | (0.38) | (0.29) | (0.26) | (0.20) | (0.18) | (0.15) | (0.13) | (0.13) | (0.13) | (0.12) | (0.12) |
| Family offenses | −0.59 | 0.15 | −0.25 | −0.18 | −0.01 | 0.20 | 0.12 | −0.01 | 0.13 | 0.15 | 0.06 | 0.11 |
| | (1.16) | (0.85) | (0.79) | (0.62) | (0.53) | (0.41) | (0.37) | (0.39) | (0.37) | (0.36) | (0.32) | (0.30) |
| Bad checks | −0.60 | 0.33 | −0.17 | −0.36 | −0.43 | −0.10 | −0.01 | −0.00 | −0.16 | −0.14 | −0.25 | −0.25 |
| | (1.67) | (0.91) | (0.65) | (0.62) | (0.63) | (0.56) | (0.50) | (0.41) | (0.38) | (0.37) | (0.30) | (0.27) |
| Liquor law | 1.39 | 1.34 | 1.21 | 0.76 | 0.19 | 0.01 | −0.33 | 0.06 | 0.00 | −0.07 | −0.11 | 0.28 |
| | (1.89) | (1.13) | (1.04) | (0.85) | (0.76) | (0.74) | (0.56) | (0.60) | (0.55) | (0.51) | (0.46) | (0.42) |
| *Inconclusive* | | | | | | | | | | | | |
| Weapons | 0.18 | 0.00 | −0.23 | −0.31 | −0.17 | −0.08 | −0.14 | −0.07 | 0.03 | 0.02 | 0.08 | 0.11 |
| | (0.50) | (0.30) | (0.27) | (0.29) | (0.18) | (0.15) | (0.13) | (0.12) | (0.12) | (0.10) | (0.09) | (0.08) |
| Arson | 4.46** | 2.06 | 2.08 | 2.74* | 2.82** | 2.31** | 2.07* | 1.82* | 2.15** | 1.65* | 1.87** | 1.93** |
| | (1.90) | (1.57) | (1.43) | (1.44) | (1.25) | (1.17) | (1.12) | (1.05) | (0.99) | (0.88) | (0.83) | (0.84) |
| Gambling | 14.04 | 6.27 | 3.18 | 3.75 | 4.63 | 3.61 | 3.06 | 2.84 | 1.78 | 0.50 | −0.56 | −0.82 |
| | (16.26) | (5.96) | (3.73) | (3.70) | (3.33) | (2.99) | (2.54) | (2.24) | (2.08) | (1.85) | (1.81) | (1.80) |
| Stolen property | 3.22 | 4.50 | 0.81 | 0.23 | −0.62 | −2.29 | −1.63 | −1.31 | −1.53 | −1.20 | −0.56 | 0.12 |
| | (7.08) | (2.78) | (2.86) | (3.07) | (2.52) | (2.23) | (2.18) | (1.97) | (1.75) | (1.48) | (1.27) | (1.16) |

*Notes.* Local-experience gradient, (near × $E$ − far × $E$) × 100, at every cumulative window, where $E = \ln(1 +$ prior same-agent same-tract listings) is the buyer's agent's local experience, under the specification of Table 5: parcel, tract × month, and agent fixed effects; standard errors clustered by tract in parentheses; $N$ = 186,510. Offense groups as in Table A4. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

# C Incident Location Classification

Several tests split an offense's incidents by *where* they occur. I classify each incident using the Chicago Police Department's `location_description` field into three mutually exclusive categories. *Commercial* covers places of business: banks, currency exchanges, and ATMs; retail, grocery, and drug stores; restaurants, bars, and hotels; gas stations; commercial, medical, and dental offices; factories and warehouses; and entertainment venues. *Residential* covers the dwelling interior and its curtilage: residences, apartments, and houses, together with garages, porches, yards, hallways, stairwells, basements, and public-housing units. *Street* covers the public way and other outdoor, encounterable space: streets, sidewalks, alleys, and gangways; parking and vacant lots; highways, parks, and waterfronts; and vehicles and transit. Incidents whose location maps to none of these, blank or ambiguous entries, are dropped, roughly one in eight. The street category is the encounterable bucket, because it is the space a passer-by can observe during a visit to the block.

This classification underlies the incident-location split (Table A5).

Applying the same split to the four priced offenses (Table A5) localizes the discount differently across them. A component significant on its own is not necessarily distinguishable from the others, so I report the Wald tests that the residential component equals the street and commercial components. The homicide and narcotics discounts are significant for incidents on the street (in 4 and 5 of 12 windows) but not for incidents at residential addresses, although the two estimates cannot be statistically distinguished; the vandalism discount is concentrated at residential addresses (8 of 12 windows) and differs significantly from street vandalism in 7 of 12 ($p = 0.018$ at the main window), with a post-sale placebo indistinguishable from zero. Burglary's near–far discount does not localize cleanly in this coarse residential–street–commercial split: neither its pooled residential nor its street component is distinguishable from the other, and the pooled residential coefficient is matched window-for-window by post-sale crime. The finer within-residential split in Table 3 is sharper: burglary's visible-*frontage* component is large and significant ($-0.74$, s.e. 0.33, significant in 7 of 12 windows) with a clean placebo ($-0.20$, $p = 0.45$), while the hidden interior is null ($-0.14$, s.e. 0.16), so the pooled residential placebo failure traces to the interior subcategory, not to the frontage incidents that carry the effect. Several additional offenses show negative residential coefficients in isolation, but those patterns are mirrored by post-sale crime and disappear when the pre-sale effect is compared directly with its placebo; the priced set is therefore not an artifact of the binary $p < .05$ placebo screen. Table A7 reports the split for every estimable offense (the priced-four summary is Table A5).

Because the reported estimates fix each offense at a single main window, Table OA11 reports the cross-location Wald tests at every exposure window, so the location heterogeneity can be read against the full window range rather than a selected one. The residential-versus-street difference for vandalism is significant across seven consecutive windows; homicide crosses conventional significance only at the twelve-month window, where its residential estimate is wrongly signed; burglary's only significant contrasts are residential-versus-commercial at ten and eleven months, driven by a residential component that fails the post-sale placebo; and narcotics never rejects. The vandalism

result is therefore not a selected-window artifact, while the street concentration for homicide and narcotics is descriptive rather than a demonstrated cross-location difference.

Table OA11: Cross-Location Wald Tests by Exposure Window, Priced Offenses

| Window | Homicide | Narcotics | Vandalism | Burglary |
|---|---|---|---|---|
| *Panel A. Residential = street/public (p)* | | | | |
| 1 | 0.227 | 0.447 | 0.495 | 0.216 |
| 2 | 0.099* | 0.647 | 0.270 | 0.520 |
| 3 | 0.526 | 0.520 | 0.238 | 0.916 |
| 4 | 0.624 | 0.948 | 0.182 | 0.712 |
| 5 | 0.678 | 0.647 | 0.090* | 0.925 |
| 6 | 0.410 | 0.843 | **0.018**** | 0.810 |
| 7 | 0.297 | 0.853 | 0.002*** | 0.425 |
| 8 | 0.136 | 0.679 | 0.001*** | **0.289** |
| 9 | 0.106 | 0.851 | 0.002*** | 0.340 |
| 10 | **0.137** | **0.897** | 0.001*** | 0.317 |
| 11 | 0.080* | 0.879 | 0.002*** | 0.340 |
| 12 | 0.046** | 0.677 | 0.000*** | 0.287 |
| *Panel B. Residential = commercial (p)* | | | | |
| 1 | 0.380 | 0.148 | 0.605 | 0.847 |
| 2 | 0.147 | 0.689 | 0.383 | 0.273 |
| 3 | 0.498 | 0.411 | 0.417 | 0.545 |
| 4 | 0.689 | 0.865 | 0.551 | 0.227 |
| 5 | 0.475 | 0.625 | 0.475 | 0.294 |
| 6 | 0.420 | 0.826 | **0.258** | 0.158 |
| 7 | 0.500 | 0.960 | 0.233 | 0.362 |
| 8 | 0.345 | 0.845 | 0.252 | **0.251** |
| 9 | 0.817 | 0.975 | 0.234 | 0.121 |
| 10 | **0.916** | **0.871** | 0.113 | 0.034** |
| 11 | 0.717 | 0.861 | 0.154 | 0.032** |
| 12 | 0.379 | 0.550 | 0.142 | 0.104 |

*Notes.* Wald $p$-values that the residential near–far component equals the street/public component (Panel A) and the commercial component (Panel B), at every exposure window 1–12 months before the sale, for the four priced offenses. Pooled 2007–2025, parcel and tract×month fixed effects, standard errors clustered by tract; each offense's main window from Table 2 is in **bold**.
$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table OA12: Residential Location Codes and Their Frontage/Interior Assignment

| CPD location code | Incidents | Main | Strict |
|---|---|---|---|
| Residence | 1,401,849 | Interior | Interior |
| Apartment | 1,032,507 | Interior | Interior |
| Residence-Garage | 134,912 | Frontage | Frontage |
| Residence Porch/Hallway† | 124,149 | Frontage | Dropped |
| Residential Yard (Front/Back) | 75,130 | Frontage | Frontage |
| Cha Apartment | 40,920 | Interior | Interior |
| Driveway - Residential | 25,198 | Frontage | Frontage |
| Cha Hallway/Stairwell/Elevator | 25,021 | Frontage | Dropped |
| Residence - Porch / Hallway | 19,974 | Frontage | Dropped |
| Residence - Yard (Front / Back) | 17,113 | Frontage | Frontage |
| Residence - Garage | 16,299 | Frontage | Frontage |
| House | 740 | Interior | Interior |
| Porch | 415 | Frontage | Dropped |
| Yard | 342 | Frontage | Frontage |
| Hallway | 111 | Frontage | Dropped |
| Garage | 75 | Frontage | Frontage |
| Cha Hallway | 40 | Frontage | Dropped |
| Vestibule | 28 | Frontage | Dropped |
| Stairwell | 28 | Frontage | Dropped |
| Cha Stairwell | 11 | Frontage | Dropped |
| Cha Lobby | 9 | Frontage | Dropped |
| Trailer | 4 | Interior | Interior |
| Cha Breezeway | 3 | Frontage | Dropped |
| Coach House | 3 | Interior | Interior |
| Rooming House | 2 | Interior | Interior |

*Notes.* Every residential-address value of the Chicago Police Department `location_description` field, its total incident count over 2001–2025, and its assignment in the within-residential frontage/interior split (Table 3). *Main* is the baseline definition: frontage (the visible outdoor curtilage and building-exterior/entry codes) versus the dwelling interior. *Strict* restricts frontage to outdoor curtilage only (garage, yard, driveway, gangway) and drops the fused porch/hallway code and interior common areas; estimates under the strict definition are not reported.

†The fused `PORCH/HALLWAY` code combines a visible porch with a hidden hallway; it is counted as frontage in the main definition and dropped in the strict definition.

The location split could also reflect objective risk: residential and street incidents may differ in how well they predict future crime. Table A8 regresses future near-ring crime on past surrounding crime separately by incident location, without fixed effects, as a buyer would see it at the sale. For vandalism, the two forecast future near-ring crime about equally well at six months ($R^2 = 0.148$ for residential and 0.125 for street), yet only residential vandalism is priced. The pricing difference therefore does not follow from a difference in recurrence, although I do not formally test the small gap in $R^2$. For homicide and narcotics, street incidents forecast better, and for burglary residential incidents do, consistent with where these offenses occur.

# D Target of the Incident

Table 3, Panel B holds observability roughly fixed (both targets are frontage incidents, visible from the street) and varies what the offense harms, splitting vandalism into damage to the property and damage to a vehicle. Table A9 reports the full twelve-window version under both fixed-effect designs. Under parcel and tract×year-month fixed effects the damage-to-property arm is negative and significant from the seventh exposure month ($-0.89$ to $-1.03$) while the damage-to-vehicle arm is null at every window, and the two differ most at the twelve-month window ($p = 0.057$). The two detailed police offense codes forecast their own recurrence at the same small rate (the Table A10 design applied to each; $\rho$ between $-0.00$ and $-0.02$, indistinguishable, $z < 1$), so the contrast holds the recurrence benchmark fixed and varies only what is harmed. Under parcel×era fixed effects both arms are negative and of similar size and are not distinguishable at any window; this specification drops to about half the sample ($N$ = 123,133), because parcel×era is identified only from parcels that resell within the same era, so it is lower-powered and its flat arms are consistent with, not evidence against, the parcel-and-month split.

Table OA13: Within-Property Visibility, All Estimable Offenses (Window-by-Window)

| Offense | Frontage $\delta$ [min, max] | Fr. sig$^-$ (plac.) | Interior $\delta$ [min, max] | Int. sig$^-$ | Fr. $\neq$ Int. |
|---|---|---|---|---|---|
| Burglary | [−0.76, −0.37] | 7/12 (0) | [−0.30, −0.11] | 0/12 | 1/12 |
| Narcotics | [−7.45, −2.35] | 7/12 (0) | [−1.20, 0.92] | 0/12 | 6/12 |
| Vandalism | [−0.60, 1.02] | 3/12 (0) | [−0.79, −0.34] | 9/12 | 3/12 |
| Homicide | [−23.69, 2.04] | 0/12 (2) | [−0.65, 6.31] | 0/12 | 0/12 |
| Larceny/theft | [−0.09, 0.17] | 0/12 (0) | [−0.08, 0.45] | 0/12 | 0/12 |
| Motor-vehicle theft | [0.52, 2.33] | 0/12 (0) | [1.02, 5.35] | 0/12 | 0/12 |
| Robbery | [−2.41, 1.90] | 0/12 (0) | [0.57, 3.89] | 0/12 | 5/12 |
| Assault (non-dom.) | [−0.43, 0.34] | 0/12 (0) | [−0.49, 0.02] | 1/12 | 0/12 |
| Assault (dom.) | [−1.14, 1.89] | 0/12 (6) | [−0.24, 0.12] | 0/12 | 0/12 |
| Sex offenses | [−4.27, 1.98] | 0/12 (0) | [−0.61, 0.12] | 1/12 | 0/12 |
| Prostitution | [−9.87, 10.85] | 0/12 (6) | [0.62, 3.31] | 0/12 | 0/12 |
| Disorderly conduct | [−0.81, 3.83] | 0/12 (0) | [−2.07, −0.98] | 2/12 | 0/12 |
| Trespass | [−0.79, 0.03] | 1/12 (0) | [−0.81, −0.28] | 0/12 | 0/12 |
| Weapons | [−10.15, −1.48] | 0/12 (0) | [−3.67, 2.33] | 0/12 | 1/12 |
| Arson | [−10.05, 0.95] | 0/12 (0) | [−5.58, 4.33] | 0/12 | 0/12 |
| Fraud | [−4.24, −1.07] | 0/12 (0) | [−0.41, −0.18] | 5/12 | 0/12 |
| Forgery | [−2.08, 10.26] | 0/12 (1) | [−1.11, 0.35] | 0/12 | 0/12 |
| Bad checks | [−1.29, 15.59] | 0/12 (5) | [0.27, 1.65] | 0/12 | 0/12 |
| Family (non-violent) | [−0.98, 6.86] | 0/12 (0) | [0.75, 2.59] | 0/12 | 0/12 |
| Gambling | [−19.36, 13.53] | 0/12 (0) | [−35.03, 30.37] | 1/12 | 0/12 |
| Stolen property | [−6.42, 16.84] | 0/12 (0) | [−17.80, 1.04] | 0/12 | 0/12 |
| Liquor law | [−20.27, 10.80] | 1/12 (8) | [−0.25, 7.95] | 0/12 | 1/12 |
| *All offenses (pooled)* | [−0.23, 0.01] | 5/12 (0) | [−0.14, 0.07] | 4/12 | 0/12 |

*Notes.* Window-by-window companion to Table 3: residential incidents of each offense split between the visible frontage (garage, porch, yard, driveway, porch/hallway and common areas; Table OA12) and the hidden dwelling interior, estimated jointly, pooled 2007–2025, parcel and tract×month fixed effects, clustered by tract. For each offense the frontage and interior columns give the range of $\delta = (\beta_{\text{near}} - \beta_{\text{far}}) \times 100$ across the twelve exposure windows; "sig$^-$" counts windows with a significant negative coefficient at five percent, and the parenthetical is the number in which the post-sale placebo (crime after the sale) is significant. The last column counts windows in which frontage differs from interior at five percent (Wald). Frontage narcotics is a small subsample (incidents recorded at the porch or garage), hence its large per-incident magnitudes and standard errors.

Table OA14: Relevance: What the Offense Threatens and Whether It Is Capitalized

| Offense | What is threatened | Capitalized |
|---|---|---|
| *Anchored to the address (stays with the property and its immediate surroundings)* | | |
| Burglary | dwelling | Yes |
| Homicide[‡] | place (violent event) | Yes |
| Narcotics | place (environment) | Yes |
| Vandalism | structure/land | Yes |
| Disorderly conduct | place (environment) | Suggestive |
| Arson | structure/land | No |
| Gambling | place (environment) | No |
| Liquor | place (environment) | No |
| Prostitution | place (environment) | No |
| Trespass[†] | structure/land | No |
| Weapons | place (environment) | No |
| *Harms a person, money, or a movable item* | | |
| Sex offenses | person | Suggestive |
| Assault (domestic) | person | No |
| Assault (non-dom.) | person | No |
| Bad checks | financial | No |
| Family | person | No |
| Forgery | financial | No |
| Fraud | financial | No |
| Larceny/theft | portable asset | No |
| Motor vehicle theft | portable asset | No |
| Robbery | person | No |
| Stolen property | portable asset | No |

*Notes.* Offenses classified by whether the harm stays with the address (dwelling, structure or land, or the place itself) or harms a person, money, or a movable item. "Capitalized" follows the offense sort in Appendix Table A3: *Yes* for the priced offenses, *Suggestive* for the suggestive tier, *No* otherwise.
[‡]Homicide is person-directed; it is placed here because its discount is statistically significant only for incidents in public space near the home, not for incidents at the dwelling. Were it classified as travelling with the household, it would be the one priced offense in that group.
[†]Trespass is imprecisely estimated: no discount is detected, but the estimate is consistent with a discount as large as 0.2 to 0.3 percent per incident.

# E Additional Price-Formation Tables

This appendix collects supporting tables for Section 5.3. Table OA15 decomposes the asking-price experience gradient into the slope among experienced agents and the first-time-in-tract differential.

Table OA15: Experience Gradient vs First-Timer Discontinuity

| Offense | Fixed effects | $\gamma_E$ (experienced slope) | $\gamma_0$ (first-timer) |
|---|---|---|---|
| Narcotics | No agent FE | −0.12 (0.07) | −0.02 (0.10) |
| Narcotics | Agent FE | −0.05 (0.05) | 0.02 (0.07) |
| Vandalism | No agent FE | −0.04 (0.03) | $0.18^{***}$ (0.07) |
| Vandalism | Agent FE | −0.01 (0.03) | $0.14^{**}$ (0.06) |
| Homicide | No agent FE | 0.09 (0.74) | 1.75 (1.61) |
| Homicide | Agent FE | −0.21 (0.56) | 1.58 (1.25) |

*Notes.* Asking-price experience gradient with a first-time-in-tract dummy (40.8% of listings), at the 12-month window, per unit of $E$. $\gamma_E$ is the slope among agents with prior experience; $\gamma_0$ is the first-timer differential in the near–far ask. For narcotics $\gamma_0$ is null, but $\gamma_E$ attenuates and is not significant once the first-timer differential is allowed for. For vandalism $\gamma_E$ collapses to zero and $\gamma_0$ is positive and significant: first-time agents ask *more* (a discontinuity, not experience accumulating). Homicide is too sparse to separate the two. Standard errors in parentheses.
$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table OA16: The Quantity Margin: Listings and Sales

| Offense | P(listed) (pp) | P(sold) (pp) | Difference (pp) |
|---|---|---|---|
| *Positive difference (listing response exceeds completed-sale response)* | | | |
| Narcotics | $0.02^{***}$ | 0.00 | $0.03^{***}$ |
| | (0.01) | (0.00) | (0.00) |
| Burglary | $0.06^{***}$ | $0.02^{***}$ | $0.04^{***}$ |
| | (0.01) | (0.01) | (0.01) |
| Arson | $0.13^{**}$ | −0.01 | $0.14^{***}$ |
| | (0.05) | (0.04) | (0.04) |
| Assault (domestic) | 0.00 | $-0.02^{***}$ | $0.02^{***}$ |
| | (0.01) | (0.00) | (0.01) |
| Homicide | $0.10^{**}$ | −0.04 | $0.14^{***}$ |
| | (0.05) | (0.04) | (0.04) |
| Vandalism | $0.02^{***}$ | 0.01 | $0.02^{***}$ |
| | (0.01) | (0.01) | (0.01) |
| Assault (non-dom.) | 0.01 | −0.01 | $0.01^{***}$ |
| | (0.01) | (0.01) | (0.00) |
| Weapons | $0.03^{*}$ | −0.01 | $0.04^{***}$ |
| | (0.02) | (0.01) | (0.01) |
| Gambling | $0.15^{**}$ | −0.02 | $0.17^{***}$ |
| | (0.07) | (0.05) | (0.06) |
| Disorderly conduct | $0.08^{***}$ | 0.04 | $0.04^{**}$ |
| | (0.03) | (0.03) | (0.02) |
| *No difference* | | | |
| Trespass | 0.01 | 0.00 | 0.01 |
| | (0.02) | (0.01) | (0.01) |
| Liquor law | 0.00 | −0.07 | 0.06 |
| | (0.07) | (0.05) | (0.06) |
| Motor-vehicle theft | −0.02 | −0.01 | −0.01 |
| | (0.02) | (0.01) | (0.01) |
| Family (non-violent) | 0.05 | 0.02 | 0.02 |
| | (0.03) | (0.03) | (0.02) |
| Forgery | 0.06 | 0.03 | 0.03 |
| | (0.04) | (0.03) | (0.03) |
| Bad checks | −0.05 | −0.01 | −0.04 |
| | (0.05) | (0.05) | (0.06) |
| Stolen property | 0.08 | 0.15 | −0.07 |
| | (0.14) | (0.12) | (0.11) |
| Fraud | −0.02 | −0.01 | −0.01 |
| | (0.01) | (0.01) | (0.01) |
| Robbery | 0.01 | 0.01 | 0.01 |
| | (0.01) | (0.01) | (0.01) |
| Larceny/theft | 0.00 | 0.00 | 0.00 |
| | (0.00) | (0.00) | (0.00) |
| Prostitution | −0.02 | $-0.02^{**}$ | 0.00 |
| | (0.02) | (0.01) | (0.02) |
| Sex offenses | −0.01 | −0.01 | 0.00 |
| | (0.02) | (0.02) | (0.02) |

*Notes.* Near–far effect on the annual probability a parcel is *listed* and that it *sells*, in percentage points (s.e. in parentheses; base rates 6.0 and 3.8 percent). P(sold) is the annual turnover rate at the parcel level, the object measured at the tract level by Boggess et al. (2013). The *difference* is listed minus sold, so a positive difference means the listing response exceeds the completed-sale response; it is not a direct withdrawal estimate (withdrawal: Table 4). Each annual parcel-year regression includes tract and community-area × month fixed effects and property controls (log building and land area, bedrooms, full and half bathrooms, year built); standard errors are clustered by tract. Parcel fixed effects are omitted so that never-listing parcels remain in the panel. $N = 13,682,888$ parcel-years. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

# F Rental Market

Table OA17: Rental Agent Local Experience (Letting Agent)

| Offense | Discount | Experience grad./SD |
|---|---|---|
| *Asking rent* | | |
| Sex offenses | $-0.21^{**}$ | $-0.048$ |
| | (0.09) | (0.065) |
| Prostitution | $-0.15$ | $-0.012$ |
| | (0.11) | (0.072) |
| Liquor law | $-0.27^{***}$ | $-0.021$ |
| | (0.10) | (0.059) |
| Burglary | $-0.11^{*}$ | 0.047 |
| | (0.06) | (0.035) |
| Narcotics | 0.10 | $-0.035^{*}$ |
| | (0.08) | (0.019) |
| *Days on market* | | |
| Sex offenses | 1.07 | $-0.085$ |
| | (0.75) | (0.632) |
| Prostitution | $-2.00^{***}$ | 0.922 |
| | (0.69) | (1.029) |
| Liquor law | 1.06 | $-0.011$ |
| | (1.81) | (0.860) |
| Burglary | 0.68 | 0.322 |
| | (0.46) | (0.451) |
| Narcotics | 1.01 | $-0.586^{***}$ |
| | (0.72) | (0.215) |
| *Withdrawal* | | |
| Sex offenses | $-0.11$ | $-0.407^{**}$ |
| | (0.32) | (0.185) |
| Prostitution | $-0.60^{*}$ | 0.325 |
| | (0.34) | (0.340) |
| Liquor law | 0.04 | 0.515 |
| | (0.43) | (0.382) |
| Burglary | $-0.04$ | 0.049 |
| | (0.19) | (0.117) |
| Narcotics | 0.10 | 0.009 |
| | (0.18) | (0.113) |

*Notes.* Near–far effect on lease listings interacted with the letting (listing) agent's local experience $E = \ln(1 + \text{prior same-agent same-tract lease listings})$, standardized. Outcomes are ln(asking rent), ln(days on market), and P(withdrawal). Parcel×era + tract×month + agent (email) fixed effects, clustered by tract; each offense at its main window. "Discount" is the near–far effect at mean experience; "Experience grad./SD" is the change per SD of $E$. Standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

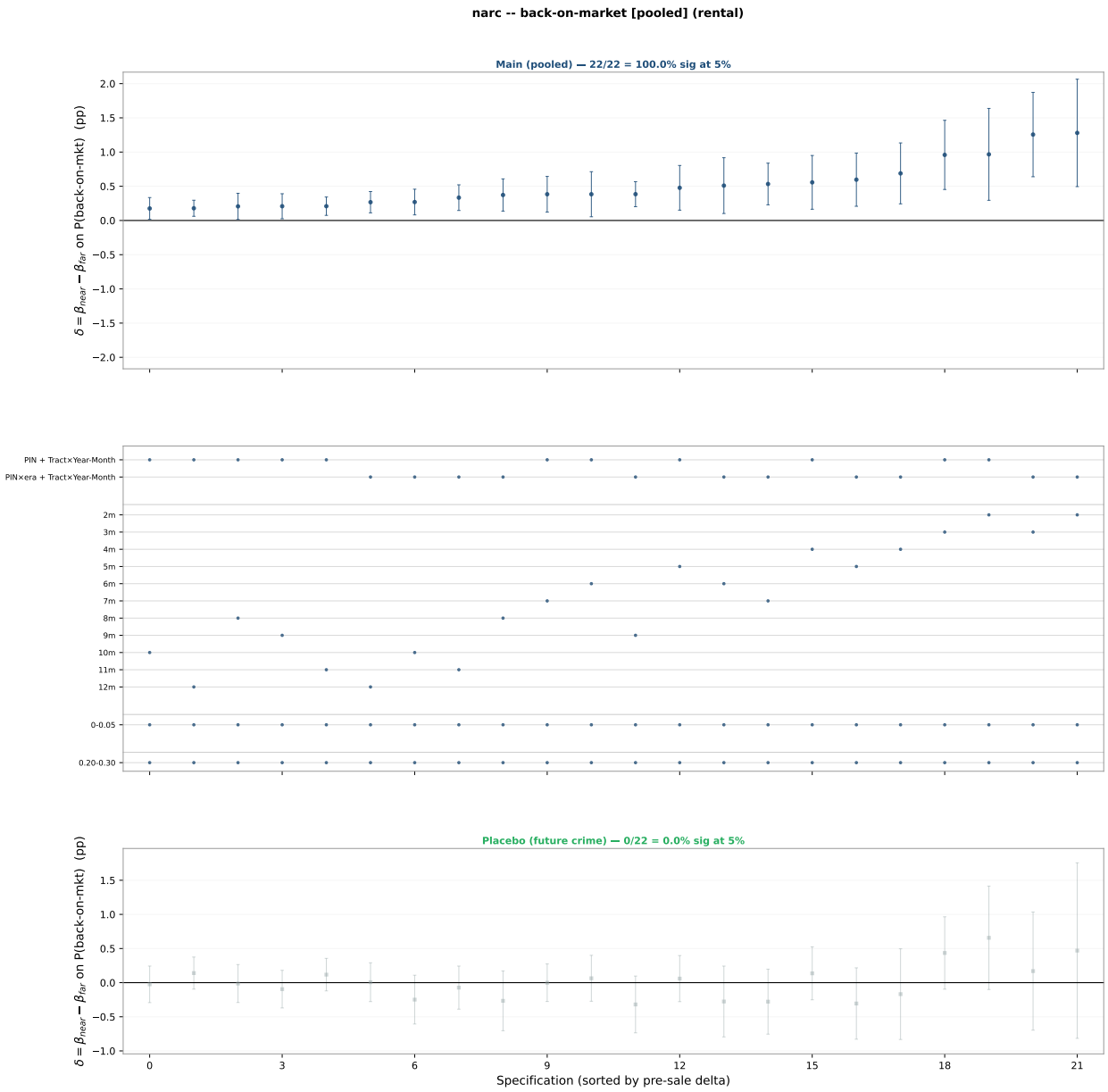



Figure OF2: Rental Leasing Friction: Relisting Response to Nearby Narcotics

*Notes.* Specification-curve "mirror" of the near–far effect ($\delta = \beta_{\text{near}} - \beta_{\text{far}}$, near 0–0.05 vs. far 0.20–0.30 mi) of nearby narcotics on the probability a lease listing is a relisting (the unit had an earlier spell on the market; percentage points), MLS lease listings, 2007–2025, clustered by tract. The top panel sorts the twenty-two specifications (eleven exposure windows, two to twelve months, × two fixed-effect designs: parcel + tract×month, and parcel×era + tract×month) by point estimate with 95% confidence intervals; the decoder panel identifies each specification; the bottom panel is the future-crime placebo (crime dated after the listing). Main: 22/22 significant at 5%; placebo: 0/22.

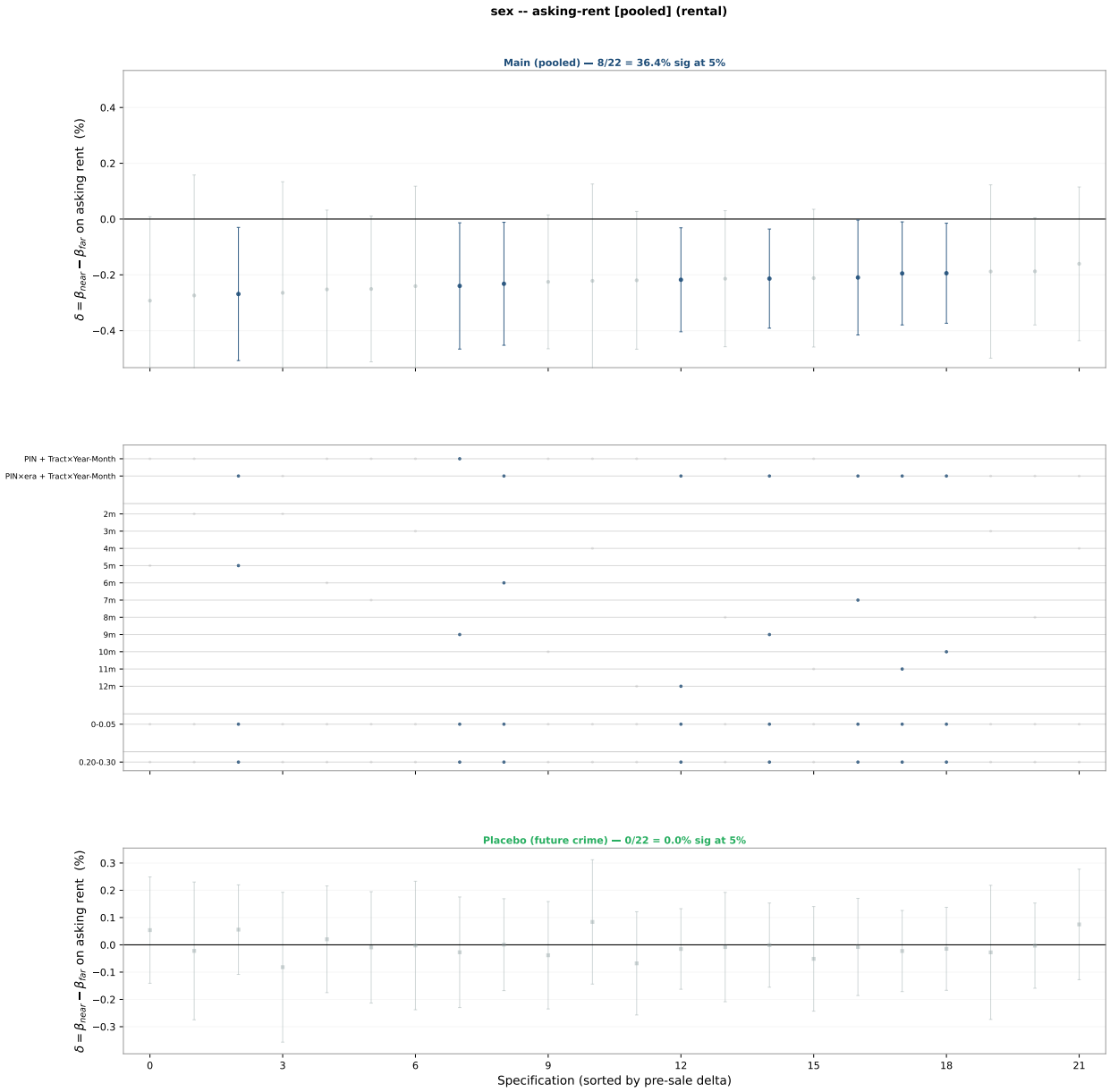



Figure OF3: Rental Asking-Rent Response to Nearby Sex Offenses

*Notes.* Specification-curve "mirror" of the near–far effect ($\delta = \beta_{\text{near}} - \beta_{\text{far}}$, near 0–0.05 vs. far 0.20–0.30 mi) of nearby sex offenses on ln(asking rent), percent, MLS lease listings, 2007–2025, clustered by tract. Twenty-two specifications (eleven exposure windows, two to twelve months, × parcel and parcel×era fixed effects), 95% confidence intervals; bottom panel is the future-crime placebo. Main: 8/22 significant at 5% (all in the discount direction); placebo: 0/22.

Table OA18: SES Heterogeneity Gradients, All Offenses

| Offense | Inc | Pov | Col | Own | HV | Rent | Blk | His | Seg | Vac |
|---|---|---|---|---|---|---|---|---|---|---|
| Homicide | $2.28^{**}$ | $-1.72$ | $3.02^{**}$ | $-0.43$ | 2.11 | $1.82^{**}$ | $-2.99^{*}$ | $-0.11$ | $-2.10$ | $-0.56$ |
| Burglary | $0.36^{***}$ | $-0.43^{**}$ | $0.52^{**}$ | 0.02 | $0.40^{**}$ | $0.30^{***}$ | $-0.61^{*}$ | $-0.40^{*}$ | $-0.43$ | $-0.13$ |
| Narcotics | $0.26^{***}$ | $-0.29^{***}$ | $0.25^{***}$ | 0.19 | $0.44^{***}$ | $0.18^{**}$ | $-0.27^{**}$ | 0.06 | $-0.15$ | 0.07 |
| Vandalism | $0.15^{**}$ | $-0.04$ | $0.31^{**}$ | $-0.17$ | 0.18 | $0.13^{**}$ | $-0.46^{**}$ | $-0.04$ | $-0.60^{***}$ | $-0.12$ |
| Theft | $0.02^{**}$ | $-0.04$ | 0.04 | $-0.02$ | 0.03 | $0.01^{*}$ | $-0.02$ | $-0.01$ | 0.01 | 0.02 |
| Motor-vehicle theft | $-0.13$ | $0.30^{**}$ | $-0.12$ | $-0.16$ | $-0.10$ | $-0.10$ | 0.17 | $-0.05$ | 0.01 | 0.00 |
| Robbery | 0.12 | $-0.34$ | 0.24 | $0.46^{**}$ | 0.03 | 0.14 | 0.05 | $-0.57^{**}$ | 0.05 | $-0.08$ |
| Assault (non-dom.) | $0.08^{*}$ | $-0.11$ | $0.21^{**}$ | 0.00 | 0.12 | $0.08^{**}$ | $-0.26^{*}$ | $-0.10$ | $-0.26^{*}$ | $-0.03$ |
| Assault (dom.) | 0.14 | 0.00 | $0.26^{**}$ | 0.03 | 0.11 | $0.12^{*}$ | $-0.15$ | $-0.16$ | $-0.09$ | $-0.03$ |
| Disorderly conduct | 0.29 | $-0.13$ | 0.48 | $-0.49$ | 0.23 | 0.29 | $-1.04$ | 0.79 | $-0.81$ | 0.18 |
| Trespass | $0.31^{***}$ | $-0.64^{***}$ | $1.02^{***}$ | 0.15 | $0.58^{***}$ | $0.19^{***}$ | $-0.95^{***}$ | $-0.74^{***}$ | $-1.06^{**}$ | 0.04 |
| Prostitution | 0.13 | $-0.09$ | 0.22 | 0.07 | 0.08 | 0.02 | $-0.25$ | 0.29 | $-0.77$ | 0.15 |
| Fraud | $0.25^{**}$ | $-0.37^{**}$ | $0.63^{***}$ | $0.20^{*}$ | $0.23^{**}$ | $0.19^{**}$ | $-1.16^{***}$ | $-0.40$ | $-0.29$ | $-0.07$ |
| Sex offenses | $-0.09$ | $-0.13$ | 0.38 | $-0.17$ | $-0.14$ | $-0.12$ | $-0.58$ | $-0.33$ | $-0.87$ | $-0.29$ |
| Forgery | 0.04 | $-0.35$ | $-0.30$ | $-0.17$ | $-0.07$ | $-0.04$ | 0.67 | 0.34 | 0.23 | 0.23 |
| Weapons | $-0.17$ | 0.03 | $-0.10$ | $-0.13$ | $-0.14$ | 0.01 | 0.13 | $-0.08$ | 0.38 | $-0.14$ |
| Arson | 0.02 | 0.45 | 1.06 | $-2.71$ | 1.86 | $-0.11$ | $-2.94$ | 2.24 | $-3.17$ | 0.71 |
| Stolen property | 1.46 | $-4.31^{***}$ | $3.16^{*}$ | $-0.25$ | $1.59^{**}$ | 1.28 | $-3.74$ | $-3.52$ | 0.95 | 0.66 |
| Liquor | $1.01^{*}$ | $-0.88$ | 1.21 | 0.36 | $1.02^{**}$ | $1.03^{**}$ | $-2.82$ | 0.09 | $-0.40$ | 0.26 |
| Gambling | $-0.18$ | $-0.86$ | $-0.60$ | 1.84 | $-1.72$ | $-0.76$ | 0.37 | $-1.17$ | 1.05 | 2.02 |
| Bad checks | $-0.27$ | 0.60 | $-0.91$ | 0.12 | 0.40 | $-0.21$ | 0.77 | 0.57 | 1.37 | 0.16 |
| Family offenses | $-0.43$ | 0.79 | $-1.13$ | $-0.73$ | $-0.43$ | $-0.01$ | 1.29 | 0.25 | $1.74^{*}$ | $1.17^{*}$ |

*Notes.* Per-SD interaction gradient (% change in the near–far discount per SD of the tract moderator) for all estimable offenses; continuous interaction, $z$ standardized over the sale sample, fully interacted, each gradient net of the tract's predetermined crime level (also interacted), parcel and tract×month fixed effects, clustered by tract. Columns: Inc=median household income, Pov=poverty rate, Col=% college-plus, Own=% owner-occupied, HV=median home value, Rent=median gross rent, Blk=% Black, His=% Hispanic, Seg=racial segregation (divergence) index, Vac=vacancy rate.

$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

# G Heterogeneity

Table OA19: Income and Racial Composition Estimated Jointly

| | Model A | | Model B | |
|---|---|---|---|---|
| Offense [window] | Income | Share Black | Income | Segregation |
| Homicide [10] | 0.86 | −2.40 | 2.06* | −1.30 |
| | (1.12) | (2.14) | (1.15) | (1.82) |
| Burglary [8] | 0.21* | −0.45 | 0.37*** | −0.34 |
| | (0.11) | (0.37) | (0.13) | (0.32) |
| Narcotics [10] | 0.21 | −0.14 | 0.41*** | 0.03 |
| | (0.16) | (0.24) | (0.13) | (0.18) |
| Vandalism [6] | 0.03 | −0.44* | 0.14 | −0.56** |
| | (0.07) | (0.24) | (0.10) | (0.22) |

*Notes.* Change in the near–far discount (percent) per standard deviation of each tract characteristic, with both characteristics interacted with the near–far term in the same regression, together with the tract's predetermined crime level. Model A: median household income and share Black (non-Hispanic). Model B: median household income and racial segregation (divergence index). Positive values mean a smaller discount. Parcel and tract × month fixed effects; standard errors clustered by tract; $N = 242{,}815$. Window in brackets (months).
$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

# H Inference and Specification Robustness

Table OA20: Arm's-Length versus Distressed Sales: Pooled Decomposition

| Offense | Window | Arm's-length (%) | Distressed (%) | $p$(diff) |
|---|---|---|---|---|
| Homicide | 10 | $-1.78^{*}$ | $-8.99$ | 0.128 |
| | | (1.08) | (4.74) | |
| Burglary | 8 | $-0.12$ | $-0.73$ | 0.254 |
| | | (0.10) | (0.52) | |
| Narcotics | 10 | $-0.00$ | $-0.64$ | 0.026 |
| | | (0.13) | (0.26) | |
| Vandalism | 6 | $-0.09$ | $-0.69$ | 0.186 |
| | | (0.07) | (0.45) | |
| Disorderly conduct | 12 | $-0.95^{***}$ | 0.58 | 0.469 |
| | | (0.36) | (2.09) | |
| Prostitution | 11 | $-0.29$ | 0.89 | 0.111 |
| | | (0.26) | (0.63) | |
| Weapons | 1 | $-0.23$ | $-5.01$ | 0.711 |
| | | (1.26) | (12.85) | |
| Arson | 4 | $-2.36$ | $-1.79$ | 0.958 |
| | | (2.56) | (10.54) | |

*Notes.* Near–far discount $\delta = (\beta_{\text{near}} - \beta_{\text{far}}) \times 100$ on ln(sale price), pooled 2007–2025, at each offense's main window (months). Arm's-length and distressed discounts come from one regression with a distress interaction; $p$(diff) is the Wald test of their equality. Arm's-length = ordinary buyer-chosen sales; distressed = REO, short-sale, or foreclosure (about 13 percent of transactions). Parcel and tract × year-month fixed effects, clustered by tract. Standard errors in parentheses; stars on the arm's-length estimate. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table OA21: Sale Prices: Alternative Standard Errors

| Crime | $\delta$ (%) | Tract | Conley 0.5 km | Conley 1.0 km |
|---|---|---|---|---|
| | | | standard error | |
| Homicide | $-3.02^{**}$ | (1.38) | (1.25) | (1.38) |
| Burglary | $-0.24^{**}$ | (0.12) | (0.11) | (0.12) |
| Narcotics | $-0.29^{**}$ | (0.14) | (0.13) | (0.17) |
| Vandalism | $-0.17^{**}$ | (0.08) | (0.07) | (0.08) |

*Notes.* Near–far close-price discount $\delta$ (%) for each priced offense, estimated on the full 2007–2025 repeat-sales sample ($N$ = 242,815) with parcel and tract×month fixed effects. Estimates are the baseline specification of Table 2 (homicide at window 10, burglary 8, narcotics 10, vandalism 6). Column 2 is the baseline standard error clustered by census tract. Columns 3–4 report Conley (1999) spatial standard errors (0.5 km and 1.0 km cutoffs, uniform kernel, great-circle distance), estimated jointly with the fixed effects, with no small-sample adjustment. Significance stars use the tract-clustered standard errors.
$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table OA22: Higher-Moment Zero-Exposure Test at the Zero-Crime Mass Point

| | Narcotics | Vandalism | Burglary |
|---|---|---|---|
| *Panel A. Variance* ($\hat{\varepsilon}^2$) | | | |
| Raw residual | 0.0002 | −0.0004 | −0.0001 |
| | (0.0007) | (0.0003) | (0.0004) |
| Winsorized (1/99) | −0.0000 | 0.0001 | −0.0001 |
| | (0.0002) | (0.0002) | (0.0002) |
| *Panel B. Skewness* ($\hat{\varepsilon}^3$) | | | |
| Raw residual | −0.0004 | −0.0015** | −0.0022*** |
| | (0.0012) | (0.0007) | (0.0009) |
| Winsorized (1/99) | 0.0001 | −0.0002 | −0.0007** |
| | (0.0004) | (0.0002) | (0.0003) |
| Observations | | 242,815 | |

*Notes.* Coefficient on the zero-exposure indicator $\mathbf{1}(\text{near} = 0)$ added to the conditioned second and third moments of the price residual (Caetano et al., 2021 higher-moment version), at the baseline PIN + tract×close-month design; standard errors clustered by tract in parentheses. A null coefficient means the moment is continuous at the mass point. The variance, the signature of heterogeneous selection, is a precise null for all three crimes. Raw-residual skewness flags for vandalism and burglary are driven by tail sales; vandalism's vanishes under winsorizing, while burglary's remains significant at the five-percent level (−0.0007). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

## H.1 Full Specification Curves (All Exposure Windows)

Figures OF4–OF7 report the sale-price specification curve for the four priced offenses across all 103 specifications. The top panel sorts the near–far discount $\delta$ (%) across the grid: 72 specifications vary the exposure window (1–12 months), the fixed-effect design (parcel + tract×month, or parcel×era + tract×month), and the far ring (0.20–0.30, 0.25–0.35, 0.30–0.40 mi) under a linear count with no controls; 24 add the full predetermined-control set at the 0.20–0.30 ring; 5 add one control at a time at the baseline specification; and 2 replace the linear count with a presence indicator or an inverse hyperbolic sine. Navy marks estimates significant and negative at five percent; maroon marks the main specification. The middle panel decodes each specification; the bottom panel is the post-sale placebo. Pooled 2007–2025, clustered by tract.

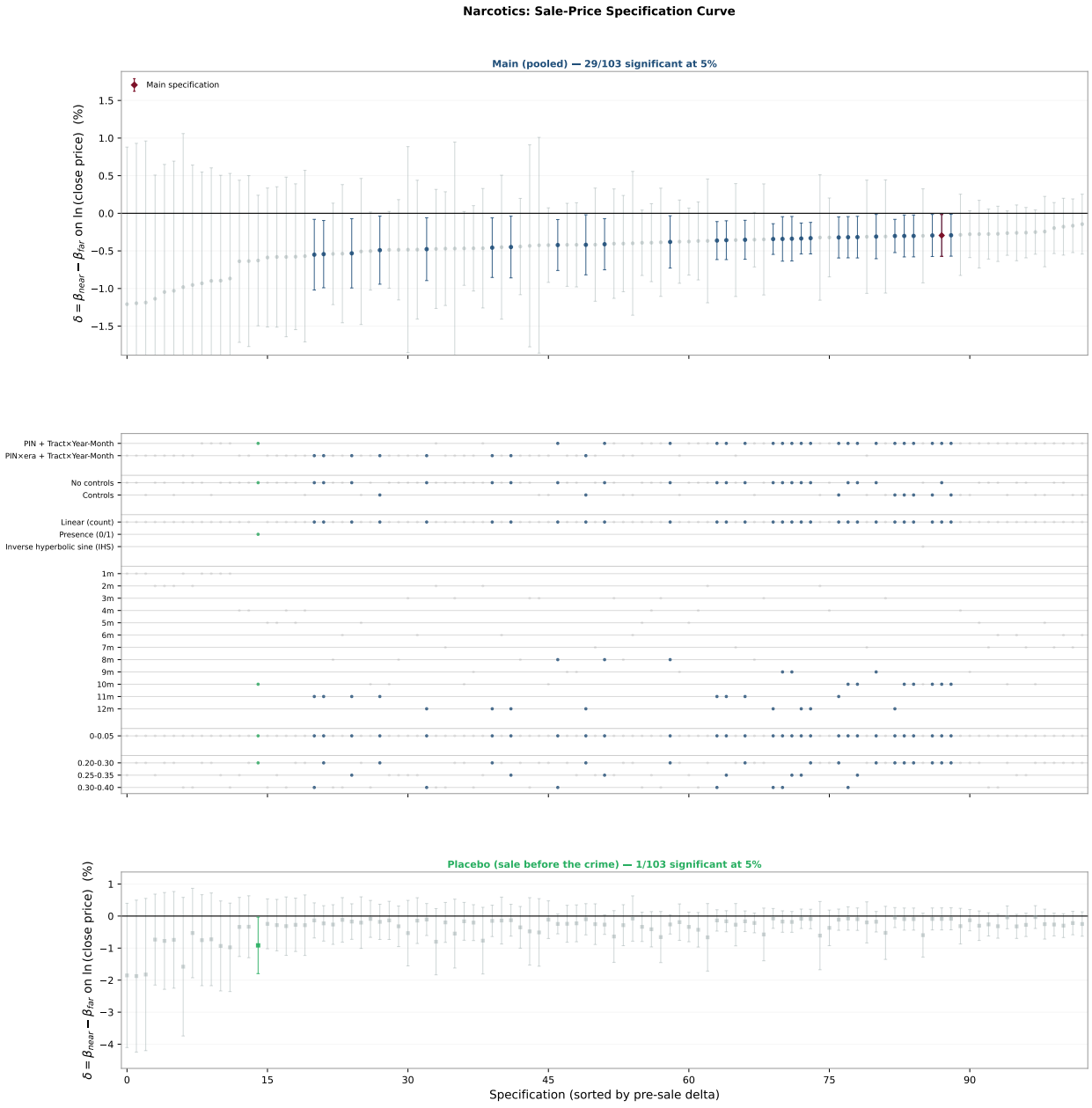



Figure OF4: Narcotics: Sale-Price Discount Across All 103 Specifications. Main: 29/103 significant; placebo 1/103.

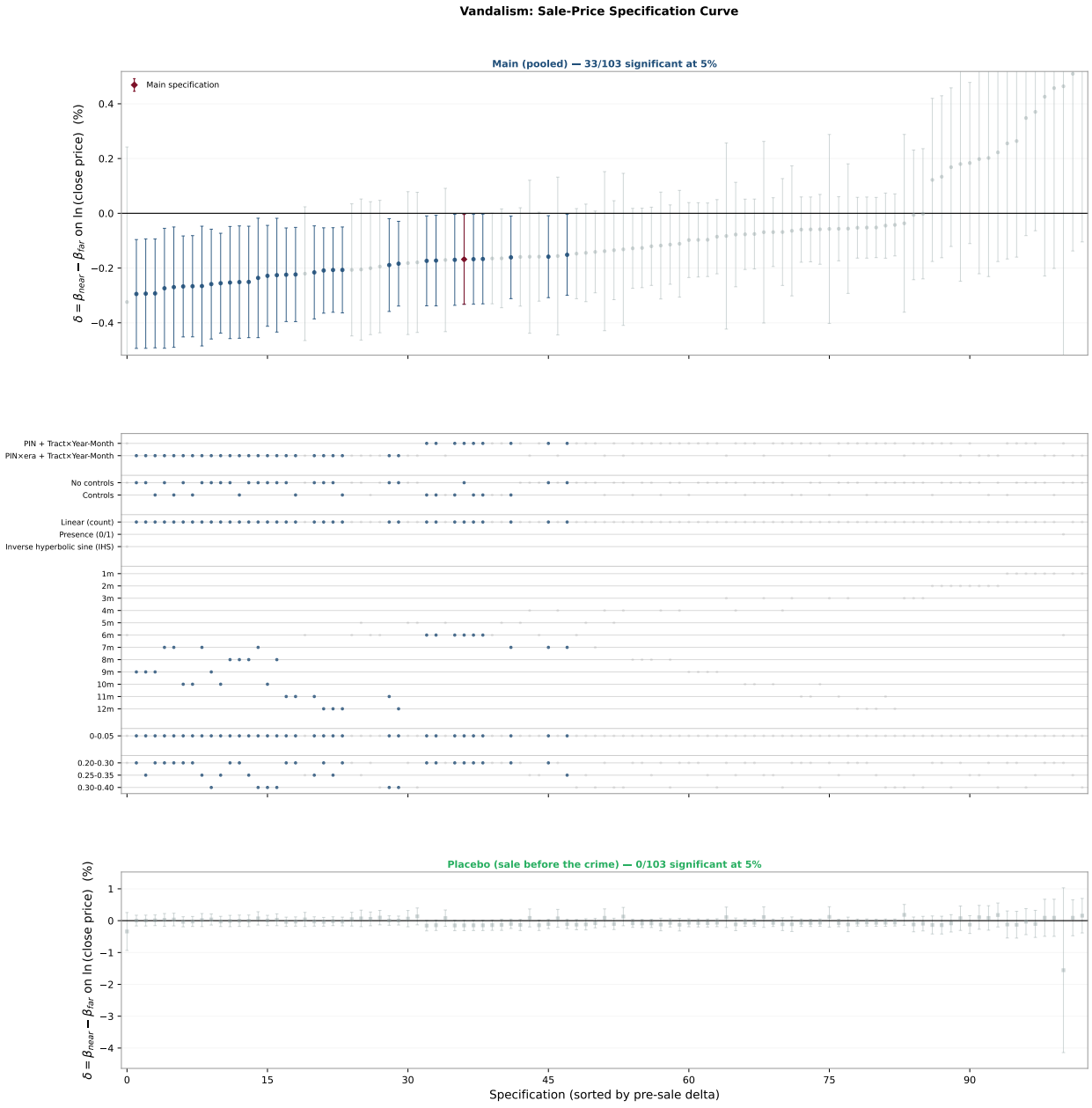



Figure OF5: Vandalism: Sale-Price Discount Across All 103 Specifications. Main: 33/103 significant; placebo 0/103.

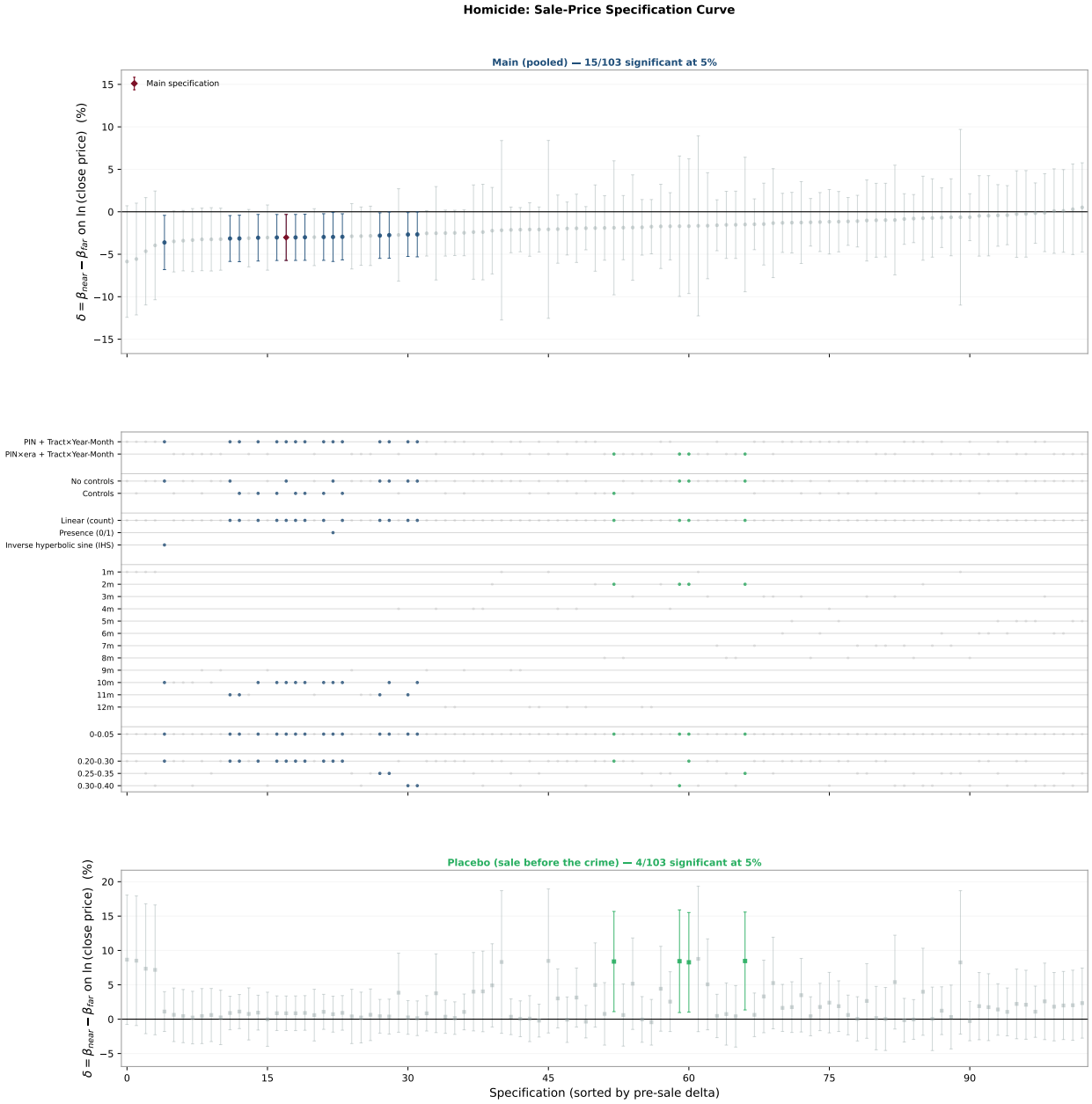



Figure OF6: Homicide: Sale-Price Discount Across All 103 Specifications. Main: 15/103 significant; placebo 4/103, all positive and at the two-month window.

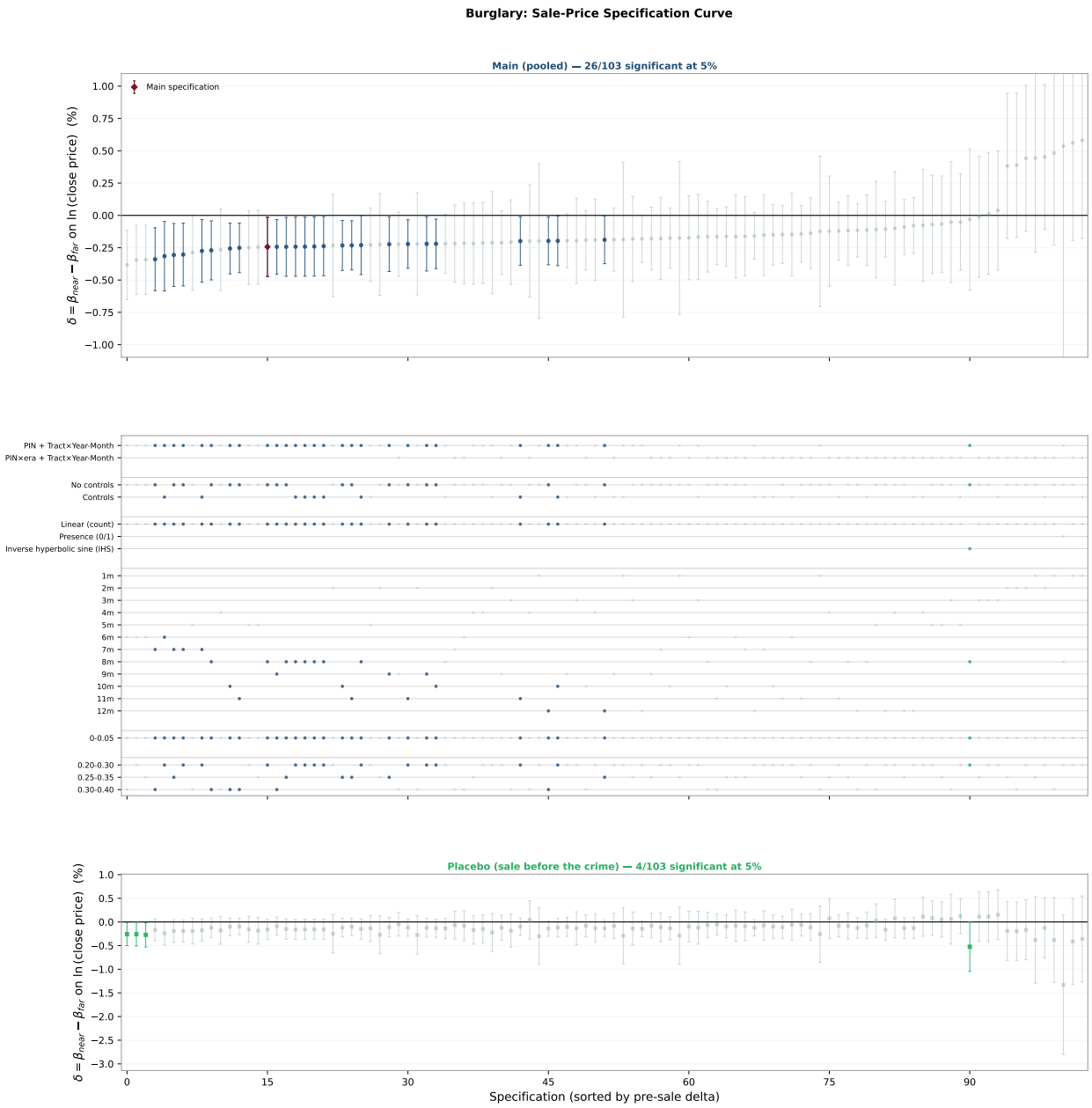



Figure OF7: Burglary: Sale-Price Discount Across All 103 Specifications. Main: 26/103 significant, all in the discount direction; placebo 4/103.

# I Functional Form, All Offenses

Table OA23: Functional Form, All Estimable Offenses

| Offense | Raw count | Presence ($\geq 1$) | IHS | Window |
|---|---|---|---|---|
| Homicide | $-3.02^{**}$ | $-2.97^{**}$ | $-3.61^{**}$ | 10 |
| | (1.38) | (1.47) | (1.63) | |
| Burglary | $-0.24^{**}$ | 0.54 | $-0.03$ | 8 |
| | (0.12) | (0.98) | (0.28) | |
| Narcotics | $-0.29^{**}$ | $-0.63$ | $-0.30$ | 10 |
| | (0.14) | (0.44) | (0.32) | |
| Vandalism | $-0.17^{**}$ | 0.46 | $-0.32$ | 6 |
| | (0.08) | (1.26) | (0.29) | |
| Theft | 0.01 | $12.01^{*}$ | $-0.15$ | 12 |
| | (0.02) | (6.92) | (0.64) | |
| Motor-vehicle theft | $-0.02$ | $-1.23$ | $-0.35$ | 12 |
| | (0.09) | (0.84) | (0.32) | |
| Robbery | $-0.03$ | $-0.35$ | $-0.25$ | 12 |
| | (0.13) | (0.74) | (0.28) | |
| Assault (non-dom.) | $-0.04$ | 0.28 | $-0.33$ | 12 |
| | (0.04) | (1.77) | (0.40) | |
| Assault (dom.) | $-0.19^{*}$ | $-0.83$ | $-0.59^{**}$ | 12 |
| | (0.10) | (1.31) | (0.28) | |
| Disorderly conduct | $-0.65^{**}$ | $-1.00^{**}$ | $-0.81^{*}$ | 12 |
| | (0.29) | (0.48) | (0.42) | |
| Trespass | $-0.14$ | $-0.66$ | $-0.52^{*}$ | 12 |
| | (0.10) | (0.59) | (0.28) | |
| Prostitution | 0.08 | 0.76 | 0.36 | 12 |
| | (0.12) | (0.98) | (0.73) | |
| Fraud | $-0.11$ | $-0.16$ | $-0.49$ | 12 |
| | (0.08) | (1.15) | (0.33) | |
| Sex offenses | $-0.47^{**}$ | $-0.66$ | $-0.74^{**}$ | 12 |
| | (0.23) | (0.47) | (0.33) | |
| Forgery | 0.21 | 0.39 | 0.26 | 12 |
| | (0.17) | (0.44) | (0.32) | |
| Weapons | 0.10 | 0.35 | 0.19 | 12 |
| | (0.28) | (0.50) | (0.50) | |
| Arson | 0.07 | $-0.43$ | $-0.12$ | 12 |
| | (1.03) | (1.08) | (1.19) | |
| Stolen property | $-0.43$ | $-0.72$ | $-0.73$ | 12 |
| | (1.05) | (1.41) | (1.50) | |
| Liquor | 0.36 | 0.95 | 0.76 | 12 |
| | (0.35) | (0.71) | (0.61) | |
| Gambling | $-1.94$ | $-3.39$ | $-3.13$ | 12 |
| | (1.75) | (2.84) | (2.47) | |
| Bad checks | $0.66^{**}$ | $1.06^{**}$ | $1.07^{**}$ | 12 |
| | (0.33) | (0.46) | (0.44) | |
| Family offenses | $1.00^{*}$ | $1.69^{**}$ | $1.69^{**}$ | 12 |
| | (0.51) | (0.71) | (0.69) | |

*Notes.* Near–far close-price effect (%) of each estimable offense's near-ring exposure entering as a raw count, a presence indicator ($\geq 1$), and inverse hyperbolic sine, each at the offense's main window, pooled 2007–2025, parcel and tract×month fixed effects, clustered by tract. The raw count is the form the dose-response implies (Appendix Table OA5). Priced offenses are discussed in the functional-form robustness check (Section 6). Standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

# J Nonparametric Distance–Time Surfaces

The near ring and exposure window in the main text are read from a nonparametric estimate of the price effect as a joint function of distance $D$ and time since incident $L$. Figure OF8 plots the estimated surface $f(D, L)$, a tensor-product spline in distance and lag, on the MLS close-date sample, 2007–2025, for the three offenses with enough incidents to identify a two-dimensional smooth: burglary, narcotics, and vandalism. Homicide is too sparse to support a stable two-dimensional surface. In every case the effect is concentrated at short distances and recent lags and flattens to zero well before the far ring, which is what justifies the near ring and the 0.20–0.30 mile far ring.

Each surface is a tensor-product spline $\text{te}(D, L)$ estimated following Korolev (2026): a signal regression of the price, demeaned by parcel and tract $\times$ month, on the crime count, with the smooth entering through a by term and grid-centered for identification, and the penalty chosen from the data by fast REML. This is a varying-coefficient model: the by term makes the crime count the treatment, and the tensor smooth lets its coefficient vary over distance and lag. The moderators are the geometry of exposure, how far the incident occurred and how recently, rather than subpopulation attributes such as buyer or neighborhood characteristics, so the surface shows how the effect of a given incident varies with the distance and recency of exposure. Following his procedure I add no wiggliness inflation, so the smoothness is what fast REML selects. Because exposure enters as counts on a discrete grid of six distance rings and twelve monthly lags rather than as continuous distances, the tensor bases are correspondingly small, six and four, below the $k = c(10, 10)$ he uses for continuous smooths. The surface is not additively separable: an additive form $f(D) + f(L)$ is rejected against the full interaction by AIC, so distance and time since incident do not enter as a sum. Multiplicative (rank-one) separability is a distinct property, and it holds only descriptively and only relative to the far-ring reference. Referenced to the untreated far ring, the same reference the near–far design assumes, the near-band surface is approximately rank-one for burglary, meaning its spatial decay has the same shape at every lag with only the amplitude fading; for narcotics it is not, the decay shape itself changing with lag. This rank-one share is descriptive, not a formal test. I therefore estimate the full tensor-product surface rather than imposing either restriction.

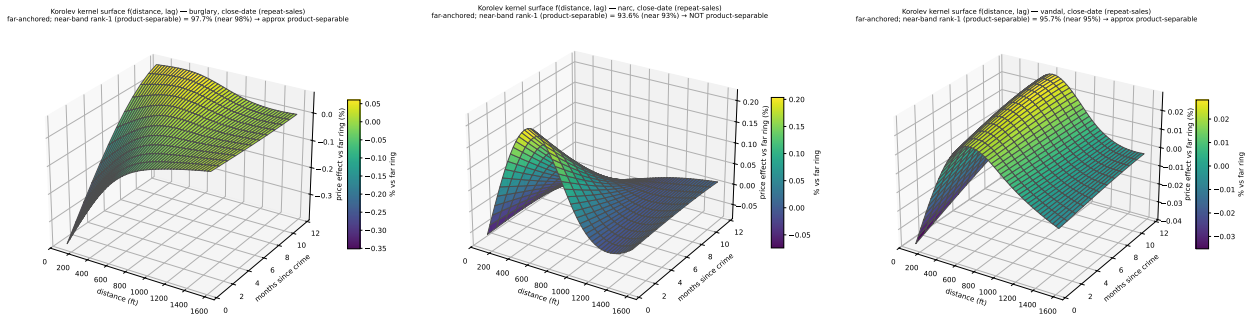



Figure OF8: Estimated price-effect surfaces $f(D, L)$ over distance and time since incident, MLS close-date sample, 2007–2025: burglary (left), narcotics (center), vandalism (right). Each is a tensor-product spline fit over the full distance domain and referenced to the far ring: its value in the 0.20–0.30 mile far ring is set to zero, consistent with the design's assumption that the far ring is untreated. The effect declines steeply in distance and lag and is close to zero before the far ring. (This far-ring reference is what makes the rank-one comparison in the text well-defined; the main-text distance profile is instead centered over its plotted range, a different reference for a different question.)

The distance profile is fit over 0–0.30 miles, the full domain spanning the far comparison ring, and centered on its average over that range, so it identifies how the effect varies with distance rather than its level, which the fixed effects absorb. The near-ring estimate is essentially invariant across the basis sizes tried, moving by only 0.011 between $c(4, 4)$ and $c(6, 12)$; the reported profile uses the $c(6, 12)$ fit, with one distance basis per 0.05-mile ring across the domain. The profile averages over the one-to-twelve-month exposure window whereas the near–far estimates are reported at each offense's main window, so the profile's homicide level is smaller than the baseline discount rather than in conflict with it; the smooth identifies the level precisely only for offenses with dense exposure, and for rarer offenses the near–far difference carries the precision.

# K Cross-Crime Confounding

A natural concern is that a discount attributed to one crime is in fact capitalizing a *different*, spatially correlated crime, that the homicide gradient, say, merely tracks nearby burglary or narcotics activity. Two facts limit this concern for homicide, burglary, and narcotics: the near-ring counts of distinct crimes are mostly weakly correlated, and, entered jointly at a common twelve-month window, each of these offenses retains its own coefficient. Vandalism is the exception.

**Near-ring counts are close to orthogonal.** Figure OF9 reports the pairwise Pearson correlation of the near-ring (0–0.05 mile) incident counts across all 923,730 MLS listings (all geocoded closed sales before the constant-parcel and repeat-sales filters), for all twenty-two non-sparse offense groups. The priced crimes are nearly independent of one another: homicide correlates at most 0.14 with any other offense group, and the strongest pair among the four priced crimes, burglary and vandalism, reaches only 0.39. Because omitted-variable bias scales with the correlation between the included and the omitted regressor, offenses that are nearly uncorrelated leave little room for one crime's gradient to be attributed to another's: even a large omitted effect, multiplied by a

correlation near 0.1, perturbs the estimated coefficient negligibly.

**A joint horse-race of all twenty-two offenses.** The mechanical argument can be checked directly. Racing only the priced crimes against one another would be selection one step later, so I enter *all twenty-two* offenses' near–far terms in a single regression (common twelve-month window, parcel and tract-by-month fixed effects, clustered by tract, $N = 242{,}815$; Table OA24). For homicide, burglary, narcotics, and disorderly conduct, the discount conditional on the other twenty-one is nearly identical to its separate estimate: narcotics $-0.29\%$ ($p = 0.01$) against $-0.33\%$ separately, burglary $-0.17\%$ against $-0.18\%$, homicide $-2.4\%$ against $-2.5\%$, and disorderly conduct $-0.58\%$. The offenses that do not price remain null when entered jointly: larceny/theft, robbery, and motor-vehicle theft stay within hundredths of a percent of zero. Standard errors are essentially unchanged, since the near-ring counts are weakly correlated. Because the horse-race is run at a common twelve-month window, it does not test vandalism's six-month discount: at twelve months vandalism's separate estimate is already small and insignificant ($-0.05$, s.e. 0.06), and its joint estimate is $-0.02$.

**Where co-movement occurs.** The two most correlated pairs in Figure OF9, larceny/theft with fraud (0.65) and larceny/theft with non-domestic assault (0.52), involve only offenses that do not price. Vandalism, however, co-moves with the high-volume offenses: domestic assault (0.49), non-domestic assault (0.45), and other offenses (0.41), and it correlates with burglary at 0.39. Narcotics correlates at 0.37 with domestic assault and with gambling. Homicide correlates at most 0.14 with any other offense.

**The tighter ring is the conservative choice.** Widening the near ring to 0–0.10 mile raises the cross-crime correlations, burglary and vandalism rise from 0.39 to 0.57, because a larger disk pools more of the shared spatial background. The 0–0.05 mile ring used throughout is therefore the specification under which distinct crimes are *most* separable, not the one that most helps the argument. Choosing it makes cross-crime confounding harder to sustain, not easier.

Table OA24: Cross-Crime Horse Race: Each Offense Conditional on the Other Twenty-One

| | Separate | | Joint (all 22) | |
|---|---|---|---|---|
| Offense | Discount (%) | (SE) | Discount (%) | (SE) |
| Homicide | $-2.49^{*}$ | (1.36) | $-2.44^{*}$ | (1.36) |
| Burglary | $-0.18^{*}$ | (0.09) | $-0.17^{*}$ | (0.09) |
| Narcotics | $-0.33^{***}$ | (0.11) | $-0.29^{**}$ | (0.11) |
| Vandalism | $-0.05$ | (0.06) | $-0.02$ | (0.06) |
| Disorderly conduct | $-0.65^{**}$ | (0.29) | $-0.58^{**}$ | (0.29) |
| Sex offenses | $-0.47^{**}$ | (0.23) | $-0.40^{*}$ | (0.23) |
| Assault (domestic) | $-0.19^{*}$ | (0.10) | $-0.16$ | (0.10) |
| Assault (non-dom.) | $-0.04$ | (0.04) | $-0.01$ | (0.04) |
| Robbery | $-0.03$ | (0.13) | $-0.01$ | (0.13) |
| Larceny/theft | 0.01 | (0.02) | 0.02 | (0.02) |
| Motor vehicle theft | $-0.02$ | (0.09) | $-0.00$ | (0.08) |
| Trespass | $-0.14$ | (0.10) | $-0.07$ | (0.09) |
| Fraud | $-0.11$ | (0.08) | $-0.14^{*}$ | (0.08) |
| Weapons | 0.10 | (0.28) | 0.20 | (0.28) |
| Arson | 0.07 | (1.03) | 0.15 | (1.03) |
| Prostitution | 0.08 | (0.12) | 0.11 | (0.11) |
| Forgery | 0.21 | (0.17) | 0.23 | (0.17) |
| Bad checks | 0.66 | (0.33) | 0.72 | (0.34) |
| Liquor | 0.36 | (0.35) | 0.56 | (0.37) |
| Family | 1.00 | (0.51) | 1.12 | (0.51) |
| Gambling | $-1.94$ | (1.75) | $-1.10$ | (1.85) |
| Stolen property | $-0.43$ | (1.05) | $-0.48$ | (1.03) |

*Notes.* Each offense's near–far discount $\delta = (\beta_{\text{near}} - \beta_{\text{far}}) \times 100$ (percent sale-price change per near-ring incident), at a common twelve-month exposure window, pooled 2007–2025, with parcel and tract × year-month fixed effects, clustered by tract ($N$ = 242,815, 679 clusters). The *Separate* columns estimate one offense at a time; the *Joint* columns enter all twenty-two offenses' near and far terms in a single regression, so each discount is conditional on the other twenty-one. Reported as a robustness exhibit; standard errors are essentially unchanged in the joint specification. Stars from the joint $p$-value, shown only for discounts.

$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

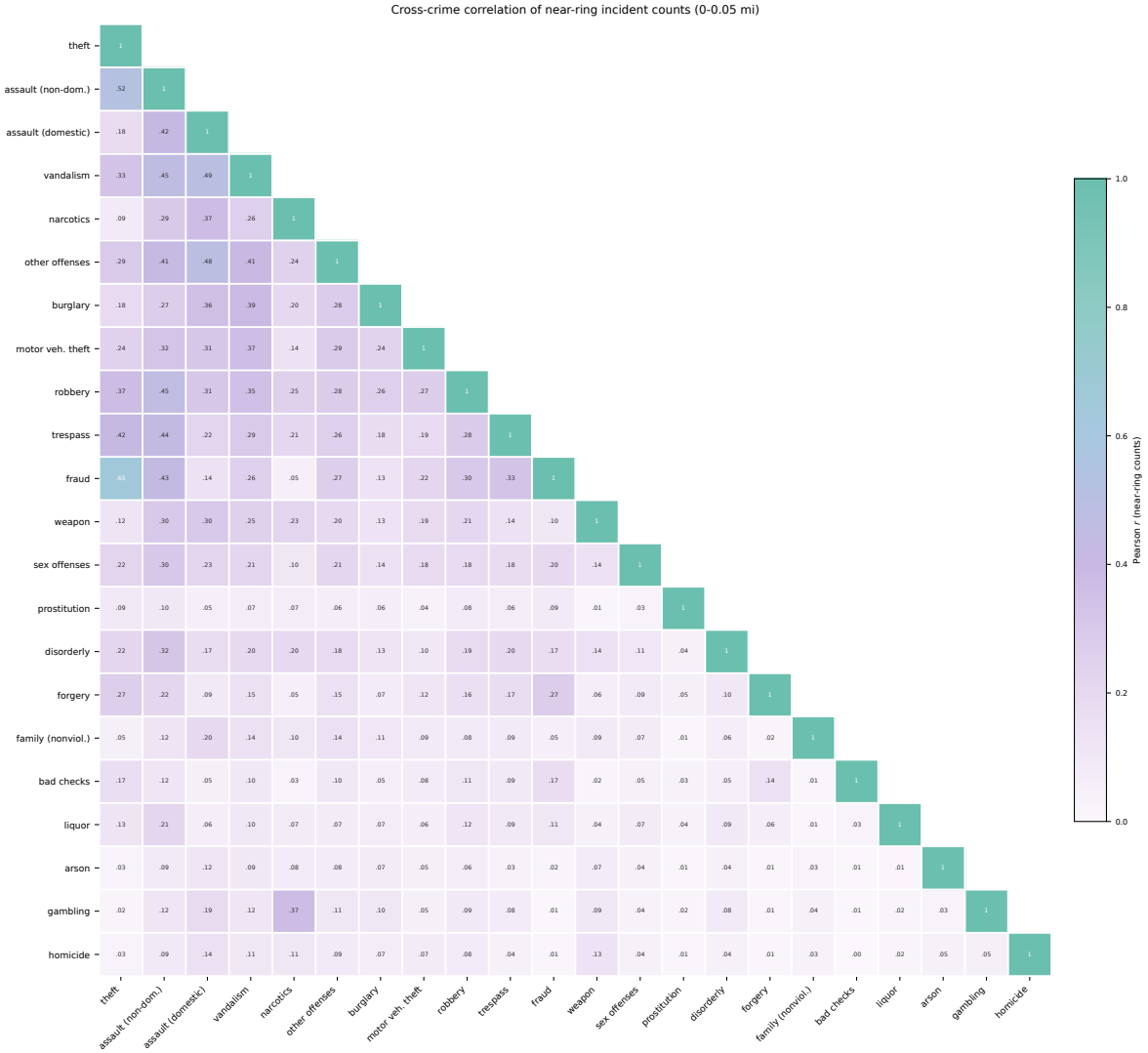



Figure OF9: Cross-crime correlation of near-ring incident counts. Pairwise Pearson correlation of the 0–0.05 mile incident counts of the twenty-two non-sparse offense groups, across 923,730 MLS listings. Homicide, burglary, and narcotics are nearly uncorrelated with the other offenses; vandalism is the exception, correlating moderately with the high-volume nuisance offenses (domestic and non-domestic assault and the "other" group) with which it co-occurs. The only strongly correlated pairs (larceny/theft–fraud, larceny/theft–assault) are among the crimes that do not price. The wider 0–0.10 mile ring raises these correlations, confirming that the 0–0.05 mile ring is the more separable specification.

# L Does Repeat-Sales Selection Drive the Discount? A Heckman Correction

The capitalization estimates are identified from repeat sales: a parcel enters the identifying sample only if it transacts at least twice, so that its fixed effect can be differenced out. This design invites an obvious objection. If nearby crime changes *which* parcels resell, the sample of twice-sold homes is endogenously selected, and the estimated discount could reflect who-sells rather than what-crime-does-to-price. I confront this directly with a Gatzlaff–Haurin two-equation selection correction (Gatzlaff and Haurin, 1997; Heckman, 1979), adapted to the MLS close-date panel.

**A selection model with an excludable instrument.** Selection operates through two transaction margins, each estimated as an annual-risk-set probit. A *first-close* probit models whether a parcel records its first MLS sale in a given year (covariates: living area, lot size, age, unemployment); a *second-close* probit models whether a previously-sold parcel resells (adding the excluded instrument). From each probit I form the inverse-Mills ratio $m = \phi(Z\pi)/\Phi(Z\pi)$ and attach it to the corresponding transaction. Because a home's first and second sales draw their $m$ from *different* probits, the selection term varies within a parcel and is not swept away by the parcel fixed effect, a subtle but essential feature that makes the correction identified in a repeat-sales design.

The excluded instrument is mortgage-rate *lock-in*: the gap between the prevailing thirty-year rate at a potential resale and the rate the owner locked in at purchase, $r_{30}(t) - r_{30}(t_{\text{prior}})$. Lock-in strongly governs *whether* an owner sells (surrendering a below-market mortgage is costly, so a large positive gap suppresses mobility), but it bears no plausible relation to the *cross-sectional near-versus-far crime gradient* realized at the moment of sale. It shifts selection without touching the price equation.

**Results.** Table OA25 delivers the sharp result. The inverse-Mills coefficient is large, positive, and significant at the 0.1% level for every priced crime: the repeat-sales sample *is* selected, and the correction has real bite. This matters because it means the test is not vacuous: the data do exhibit the selection the objection worries about, and the Heckman term is capable of moving the estimate. Adding $m$ changes the point estimates little: homicide shifts from $-4.35\%$ to $-4.18\%$, narcotics from $-0.47\%$ to $-0.52\%$, vandalism from $-0.40\%$ to $-0.34\%$, and burglary from $-0.36\%$ to $-0.33\%$; burglary is not significant either before or after the correction in this smaller sample.

The logic is what makes this convincing. A null correction is uninformative when the instrument is weak or selection is absent; one learns nothing from a test with no power. Here selection is strongly present ($m$ highly significant) and is explicitly purged, and the point estimates change little. The composition of the twice-sold sample is not random, $m$ says as much, but that selection is *orthogonal to the near-versus-far crime gradient*: it shifts the price level, not the crime slope. For homicide, narcotics, and vandalism, the discounts are therefore not explained by which homes resell; for burglary the check is uninformative because the estimate is imprecise in this sample.

Table OA25: Capitalization Under a Repeat-Sales Selection Correction

| Crime | Uncorrected $\delta$ (%) | Selection-corrected $\delta$ (%) | Selection term $m$ (Mills) |
|---|---|---|---|
| Homicide | $-4.35^{**}$ | $-4.18^{**}$ | $0.72^{***}$ |
| Burglary | $-0.36$ | $-0.33$ | $0.72^{***}$ |
| Narcotics | $-0.47^{***}$ | $-0.52^{***}$ | $0.72^{***}$ |
| Vandalism | $-0.40^{**}$ | $-0.34^{**}$ | $0.72^{***}$ |

Near = 0–0.05 mi, far = 0.20–0.30 mi; outcome ln(close price); PIN and tract×close-month fixed effects; standard errors clustered by tract; pooled sample, each offense at a fixed window (homicide 10, burglary 8, narcotics 12, vandalism 7 months), which differ from the main windows for narcotics and vandalism, with a persistence control for near-ring crime beyond the window cumulated to twelve months. $m$ is the inverse-Mills ratio from a two-probit Gatzlaff–Haurin selection model (first-close and second-close probits; excluded instrument = mortgage-rate lock-in, plus unemployment). The coefficient on $m$ is nearly identical across offenses (0.717 to 0.720) because the selection equations are driven by house characteristics, lock-in, and unemployment rather than by crime. "Uncorrected" and "selection-corrected" run on the identical $m$-matched repeat-close sample ($N$ = 130,564) for comparability, smaller and differently selected than the full repeat-sales sample of the main results, so the uncorrected homicide estimate here ($-4.35$) exceeds the baseline repeat-sales discount ($-3.0$ percent); the object of the test is the movement from uncorrected to corrected, not the level. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

# M Is Nearby Crime Exogenous? A Zero-Exposure Test

The capitalization estimates identify off within-parcel variation: a home enters the identifying sample only through repeated sales, and the near–far discount is the price movement associated with a change in nearby crime, net of the parcel and neighborhood–month fixed effects. Appendix L shows the estimate is robust to selection into the twice-sold sample. Here I test a distinct assumption, that the near-ring crime count is itself *exogenous* conditional on the design, i.e. uncorrelated with unobserved determinants of price. I do so with a zero-exposure test that requires no exclusion restriction, exploiting a feature of the data: the near-ring count has a *mass point* at zero, because most homes record no incident in the exposure window.

**The zero-exposure test.** I implement the test of Caetano et al. (2021). I augment the baseline near–far specification (Equation (5)) with a zero-exposure indicator, equal to one when the near ring records no incident, under the same fixed effects:

$$\ln(\text{close})_{it} = \beta_{\text{near}}\text{Near}_{it} + \beta_{\text{far}}\text{Far}_{it} + (\text{persistence}) + \gamma\,\mathbf{1}(\text{Near}_{it} = 0) + \mu_i + \lambda_{c(i)t} + \varepsilon_{it}, \quad (1)$$

where $\mu_i$ is the parcel fixed effect and $\lambda_{c(i)t}$ the tract×month fixed effect of Equation (5), and I test whether $\gamma = 0$. The logic, due to Caetano (2015) and applied to crime counts with a mass at zero by Caetano and Maheshri (2018) to select among dynamic-spillover models of Dallas police data, is that confounders tend to be *discontinuous* at a mass point: if the zero-crime homes lie on the

smooth dose–response traced by the crime-exposed homes, the indicator is zero; a nonzero coefficient signals that the mass-point observations are selected. This regression version dominates the original nonparametric test: it is more powerful, applies to discrete or mixed treatments, and, in a two-way fixed-effects design, jointly detects failures of exogeneity, functional form, *and* the "strong" parallel-trends condition (parallel trends plus uncorrelated treatment effects, the assumption whose failure drives the negative-weights problem).

**Which crimes can be tested.** The test needs positive support beyond the mass point, homes at one, two, three or more nearby incidents, to trace the dose–response. Homicide fails this: its near ring is $\approx 99\%$ zero, with essentially no observations above one (Figure OF10); arson and stolen property are similarly sparse (not shown), so the test is not identified for them; they are, by construction, a presence contrast rather than a dose. Narcotics, vandalism, and burglary have workable mass points (48–75% zero, with 12–27% of homes at two or more) and are tested.

**Selection into zero exposure.** Table OA26 reports the coefficient on the zero-exposure indicator as fixed effects escalate. Uncontrolled, it is large and highly significant: for narcotics, zero-crime homes sell for about 50% (a 40.5 log-point coefficient on the zero-exposure indicator) more than the dose–response predicts, and they differ so much on unobservables that the raw near–far slope even flips positive. Tract×close-month fixed effects shrink it by an order of magnitude, and under the baseline PIN + tract×close-month design it is a precise null for both narcotics and vandalism (Figure OF11). Crucially, the discount $\delta$ is essentially unchanged whether or not the zero-exposure indicator is included (Panel B). The within-property, within-neighborhood-month design thus absorbs the cross-sectional discontinuity that the raw data plainly exhibit; the diagnostics provide supporting evidence that the estimated discount is not driven by *which* homes have no nearby crime.

**Burglary.** The zero-exposure and skewness tests both indicate that zero-exposure parcels differ for burglary. Two estimates identify the slope from exposed parcels only: adding the zero-exposure indicator gives $-0.552$ (s.e. 0.168), and the binned dose-response (Appendix Table OA5) gives 0.48 (s.e. 0.25) for a single incident and $-1.85$ (s.e. 0.76) for four or more, a cumulative rather than per-incident effect. The rejection does not establish the direction of any remaining bias, so the baseline linear specification, $-0.244$ (s.e. 0.117), is an approximation, not a bound. The jump is not spillover from the adjacent ring (controlling for 0.05–0.10 burglary leaves it at $-0.99$, s.e. 0.32), and at a 0–0.10 near ring the test does not reject ($-0.06$, s.e. 0.27) while the discount remains significant ($-0.14$, s.e. 0.07).

**Higher moments.** Finally, I apply the same zero-exposure test to the second and third moments of the price residual, conditioned on the covariates (Figure OF12 and Table OA22). A discontinuity in the *variance* at the mass point would signal *heterogeneous* selection, the correlated random-effects failure that also invalidates two-way fixed effects. Adding $\mathbf{1}(\text{near} = 0)$ to the conditioned second

Table OA26: Zero-Exposure Test of Identification (Caetano–Caetano–Fe–Nielsen 2021)

| | Narcotics | Vandalism | Burglary |
|---|---|---|---|
| *Panel A. Zero-exposure indicator* $\mathbf{1}(near = 0)$*, coefficient (%)* | | | |
| Raw (no fixed effects) | $40.500^{***}$ | $4.924^{***}$ | $-1.353$ |
| | (2.830) | (1.212) | (1.176) |
| Tract × close-month | $4.314^{***}$ | $2.518^{***}$ | 0.329 |
| | (0.784) | (0.578) | (0.538) |
| PIN + tract × close-month (baseline) | $-0.129$ | $-0.323$ | $-0.978^{***}$ |
| | (0.386) | (0.256) | (0.317) |
| + quadratic dose-response | $-0.061$ | $-0.580^{*}$ | $-1.057^{**}$ |
| | (0.493) | (0.297) | (0.442) |
| *Panel B. Near–far discount* $\delta$ *(%), baseline design* | | | |
| $\delta$, without dummy | $-0.294^{**}$ | $-0.168^{**}$ | $-0.244^{**}$ |
| | (0.142) | (0.084) | (0.117) |
| $\delta$, with dummy | $-0.308^{*}$ | $-0.246^{**}$ | $-0.552^{***}$ |
| | (0.157) | (0.107) | (0.168) |
| Observations | | 242,815 | |

*Notes.* The zero-exposure test of Caetano et al. (2021) adds a zero-exposure indicator, $\mathbf{1}(\text{near} = 0)$, to $\ln(\text{close}) = \beta_1 \text{near} + \beta_2 \text{far} + \text{persistence} + \varepsilon$; under valid identification its coefficient is zero. In two-way fixed-effects designs the test jointly detects failures of exogeneity, linearity, and strong parallel trends. Near = 0–0.05 mi, far = 0.20–0.30 mi, each offense at its main window (narcotics 10, vandalism 6, burglary 8 months), with a persistence control for near-ring crime in the remaining months up to twelve; single-parcel sample; standard errors clustered by tract in parentheses. The large raw discontinuity (narcotics 40.5 log points, about 50%: zero-crime homes are dramatically different uncontrolled) is progressively absorbed by fixed effects and is a precise null under the baseline PIN + tract×close-month design; the discount $\delta$ is stable with and without the zero-exposure indicator for narcotics and vandalism. The zero-exposure and skewness tests both indicate that zero-exposure parcels differ for burglary. Two estimates identify the slope from exposed parcels only: adding the zero-exposure indicator gives −0.552 (s.e. 0.168), and the binned dose-response (Appendix Table OA5) gives 0.48 (s.e. 0.25) for a single incident and −1.85 (s.e. 0.76) for four or more, a cumulative rather than per-incident effect. The rejection does not establish the direction of any remaining bias, so the baseline linear specification, −0.244 (s.e. 0.117), is an approximation, not a bound. The jump is not spillover from the adjacent ring (controlling for 0.05–0.10 burglary leaves it at −0.99, s.e. 0.32), and at a 0–0.10 near ring the test does not reject (−0.06, s.e. 0.27) while the discount remains significant (−0.14, s.e. 0.07). Homicide, arson, and stolen property are not testable (near-ring ≈ 99% zero: too few properties with any nearby incident). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

moment yields a precise null for all three crimes (burglary $p = 0.64$, narcotics 0.99, vandalism 0.51; winsorized, at the baseline parcel and tract×close-month design, 0–0.05 mile, counts at each offense's main window): the diagnostics do not detect a variance discontinuity at zero exposure. Winsorization removes the skewness rejection for vandalism, but burglary's remains significant at the five-percent level ($p = 0.02$). For burglary, the skewness rejection is consistent with the selection at zero exposure that the zero-exposure indicator detects. Identifying the slope from exposed parcels only gives −0.552 (s.e. 0.168); because the test does not establish the direction of bias, the baseline linear specification (−0.244) is an approximation, not a bound.

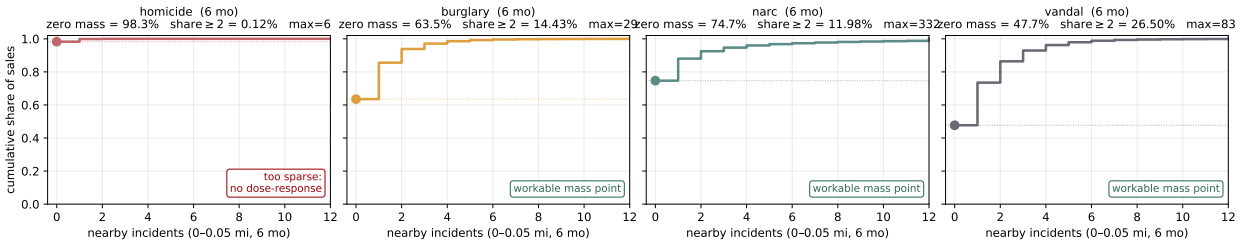



Figure OF10: Which crimes can be tested: empirical CDF of the near-ring count. The zero-exposure test needs a zero mass point *and* positive support above it. Homicide is 99% zero, too few properties with any nearby incident; narcotics, vandalism, and burglary have workable mass points.

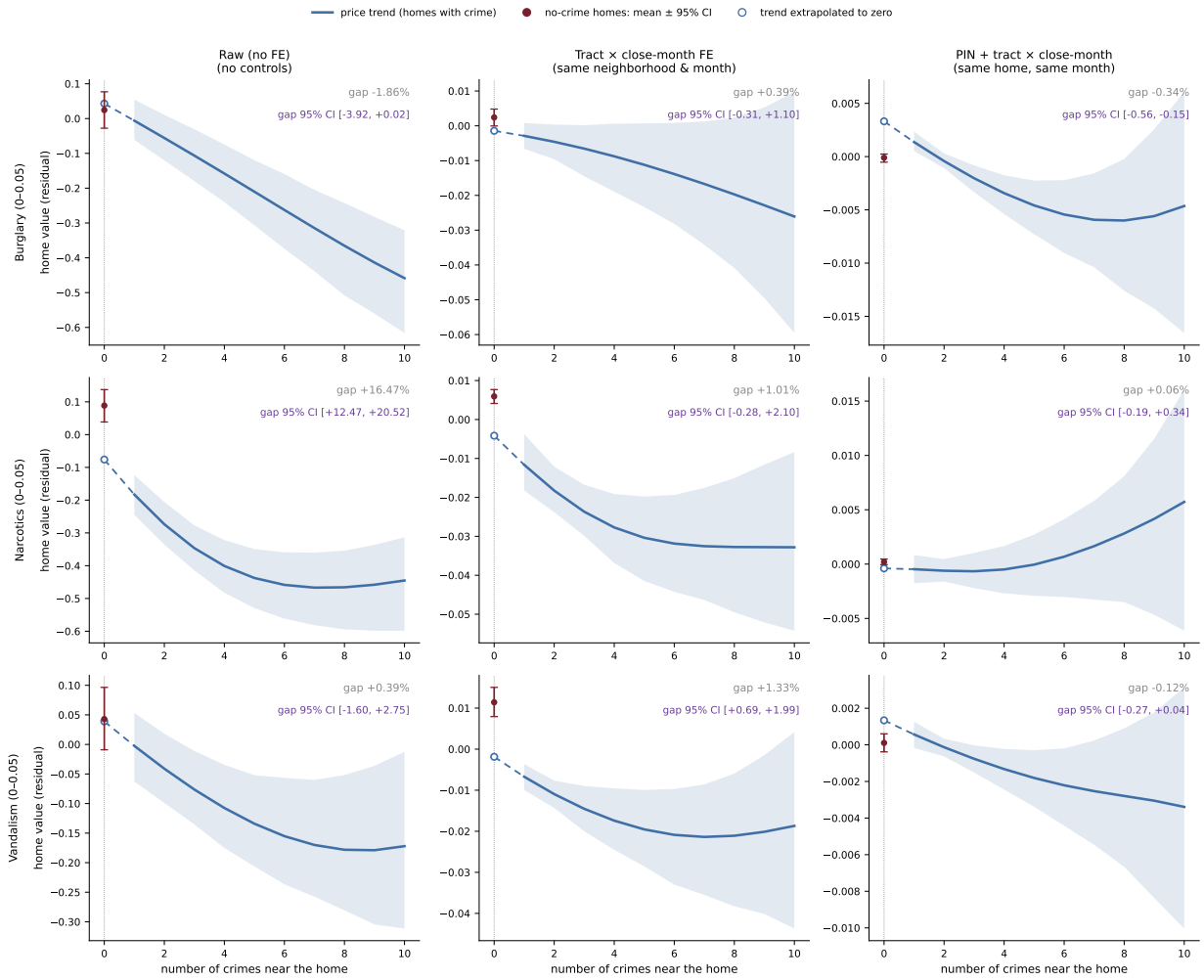



Figure OF11: The zero-exposure diagnostic. Blue: dose–response among crime-exposed homes; red: no-crime homes; open circle: the trend extrapolated to zero. For narcotics and vandalism the large gap in the raw column collapses to zero under the baseline PIN + tract×close-month design; for burglary a gap remains, so its discount is identified partly from the exposed-versus-unexposed contrast; the rejection does not establish the direction of any bias (Table OA26). With a 0–0.10 mile near ring, the burglary gap disappears (−0.06, s.e. 0.27) and the discount remains (−0.14, s.e. 0.07; Section 5.1).

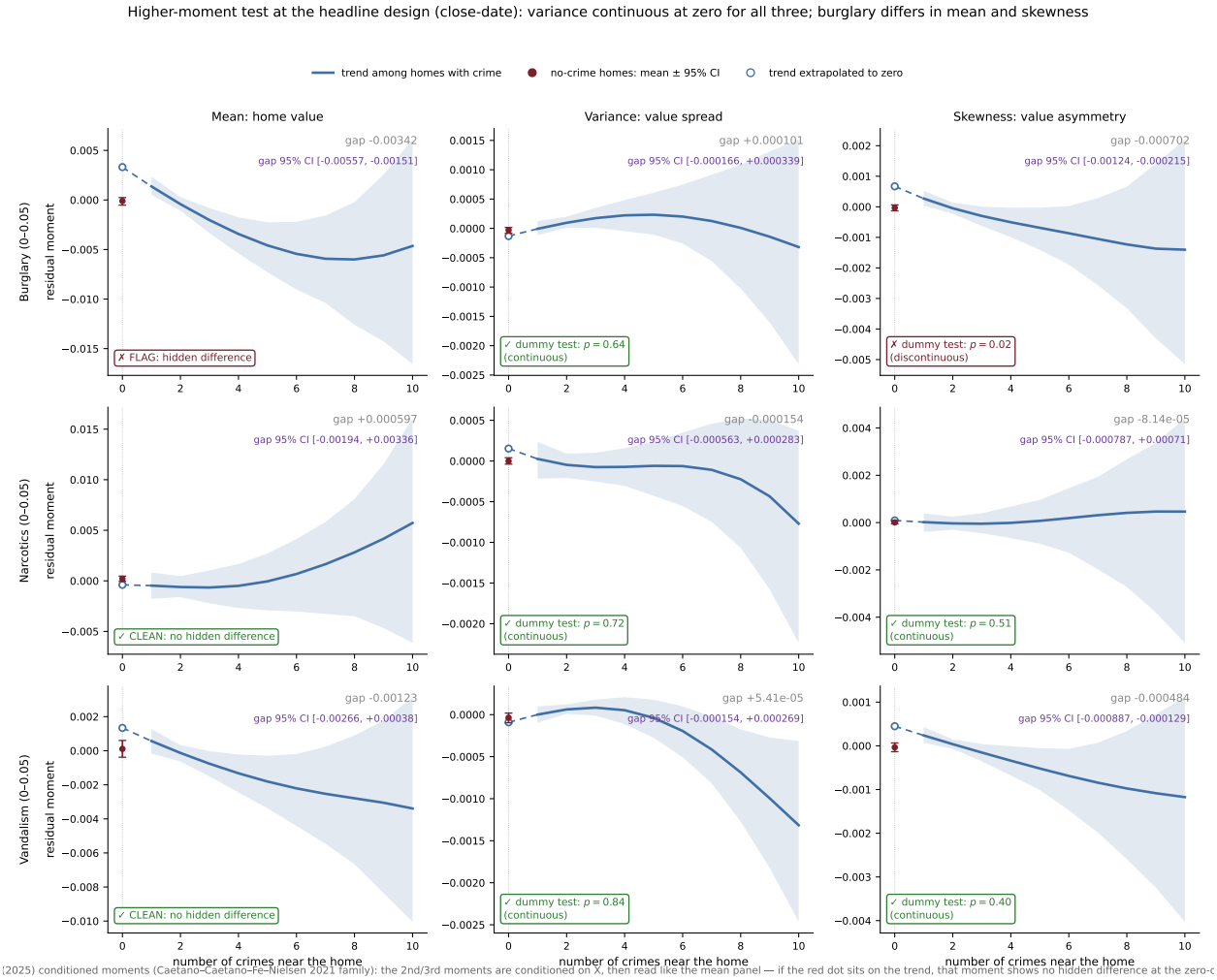



Figure OF12: Higher-moment test. The mean, variance, and skewness of the conditioned price residual at the zero-crime mass point. The parametric zero-exposure test finds the *variance*, the signature of heterogeneous selection, continuous at zero for all three crimes (burglary $p = 0.64$, narcotics 0.99, vandalism 0.51); under winsorizing the skewness rejection is removed for vandalism but not for burglary. The local-linear gaps shown are boundary-sensitive and are superseded by the zero-exposure test.

# N Market Tightness, Contract Timing, and Distressed Sales

A nearby incident can lower the recorded sale price by revising what buyers believe the home is worth, or by giving a buyer under contract leverage to reopen the agreed price. Both move the close price in the same direction, and a transaction reveals only the equilibrium price, so the baseline regression cannot separate them. I look for the signature of bargaining on the margin where a leverage account differs most sharply from revaluation, the buyer's market power.

I assign each sale the months of supply (active inventory over trailing three-month sales) in its community area in the month it was listed. Because it is measured in the month of listing, it is predetermined relative to the close. It is a standard measure of market tightness, defined within year so that a soft market is soft relative to its own year rather than to the 2009–2012 downturn. A buyer in a soft market holds outside options and leverage that a market of competing offers removes, so a bargaining channel predicts a larger discount where supply is high. Estimating the discount across terciles of supply, with the parcel and tract-by-month fixed effects and stale-crime controls of the baseline specification, I find no such gradient. For burglary, narcotics, and vandalism the market-tightness comparisons do not detect the larger discount in soft markets that bargaining predicts: at no exposure window is the discount significantly larger in soft markets; the only significant differences, for burglary at one and two months and vandalism at one month, show larger discounts in tight markets. Homicide is imprecise: at the twelve-month window its estimate is $-4.8$ percent in soft markets against $-0.3$ in tight, but the difference is not significant ($p = 0.13$) and the ordering reverses at shorter windows, so the homicide comparison is inconclusive. A tercile split defined over the whole sample period does produce an apparent gradient, but it places most soft-market sales in the 2009–2012 crisis; the gradient is an era effect that the within-year definition removes.

Two more direct tests are available but lack power in these data. The first compares crime in the twelve months before the contract date, which is embedded in the agreed price, with crime between contract and close, which arrives after the price is set and can move it only through renegotiation. Pre-contract exposure carries the usual discounts, about 0.2 percent per burglary and 0.3 percent per narcotics incident. Post-contract exposure is insignificant for every offense, but its standard errors are too wide to be informative: the burglary and narcotics estimates carry standard errors of 0.24 and 0.58, and the homicide estimate 3.3, so each interval spans effects as large as the belief-channel discount. The window between contract and close averages about six weeks, so few incidents fall in it, and holding that window to a fixed length requires restricting to slow closings, which selects troubled deals and no longer reproduces the belief channel itself. The second test compares distressed sales with arm's-length sales, but distressed transactions are about thirteen percent of the sold sample (REO, short-sale, or foreclosure, the same definition as Table OA20) and their estimates are too dispersed to sign.

The market-tightness comparisons do not show the larger soft-market discount that bargaining predicts. The short-horizon renegotiation and distressed-sale margins remain beyond the reach of these data.

## References


Lyndsay N. Boggess, Robert T. Greenbaum, and George E. Tita. Does crime drive housing sales? Evidence from Los Angeles. *Journal of Crime and Justice*, 36(3):299–318, 2013. doi: 10.1080/0735648X.2013.812976.

Carolina Caetano. A test of exogeneity without instrumental variables in models with bunching. *Econometrica*, 83(4):1581–1600, 2015. doi: 10.3982/ECTA11231.

Carolina Caetano, Gregorio Caetano, Hao Fe, and Eric Nielsen. A dummy test of identification in models with bunching. Finance and Economics Discussion Series 2021-068, Board of Governors of the Federal Reserve System, 2021.

Gregorio Caetano and Vikram Maheshri. Identifying dynamic spillovers of crime with a causal approach to model selection. *Quantitative Economics*, 9(1):343–394, 2018. doi: 10.3982/QE756.

Timothy G. Conley. GMM estimation with cross sectional dependence. *Journal of Econometrics*, 92(1):1–45, 1999.

Dean H. Gatzlaff and Donald R. Haurin. Sample selection bias and repeat-sales index estimates. *Journal of Real Estate Finance and Economics*, 14(1–2):33–50, 1997.

James J. Heckman. Sample selection bias as a specification error. *Econometrica*, 47(1):153–161, 1979.

Ivan Korolev. Estimating semiparametric and nonparametric fixed effects panel data models with `mgcv`, 2026. Working paper, arXiv:2606.12739.